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АНГЛИЙСКИЙ ЯЗЫК
ДЛЯ МАГИСТРАНТОВ
НАПИСАНИЕ НАУЧНОЙ СТАТЬИ

ENGLISH FOR MASTER STUDENTS
WRITING A RESEARCH ARTICLE

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Данное учебное пособие предназначено для магистрантов технических направлений. Пособие направлено на развитие умений академического письма, а именно обучение написанию жанра научной статьи, включающей описание основных результатов научной работы молодого исследователя. Пособие включает темы, необходимые для обучения написанию данного научного жанра и совпадающие с ее логико-семантической структурой: «Article structure. IMRaD format», «Introduction and Literature Review sections», «Materials and Methods section», «Results section», «Discussion and Conclusion Sections», «Title, Key Words and an Abstract». Данное пособие может быть использовано на очных занятиях или во время самостоятельной работы магистрантов.

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CONTENTS

Unit 1. Article Structure. IMRaD Format	4
Unit 2. Introduction and Literature Review Sections.....	9
Unit 3. Materials and Methods Section.....	17
Unit 4. Results Section.....	25
Unit 5. Discussion and Conclusion Sections.....	34
Unit 6. Title, Key Words and an Abstract.....	40
Supplementary Vocabulary and Grammar Exercises.....	47
Video Scripts	54
Appendix.....	75
References.....	77

Unit 1. Article Structure. IMRaD Format

Task 1. Answer the questions.

1. Have you ever written an article? What is the difference between an essay and an article/ a popular article and a research article?
2. How do you make a decision that an article is worth reading?
3. What is the purpose of a research article structure?
4. What is the role of an abstract?

Vocabulary

manuscript – рукопись, публикация

roadmap – карта, план

alternate – периодически изменять

showcase – демонстрировать

comprehensive – всесторонний, всеобъемлющий

Task 2. Read the text about the titles of the research article sections and say what IMRAD stands for and what the function of the Abstract section is.

Typically, research articles are written in the IMRAD format: Introduction, Materials and Methods, Results and Discussion. This format helps the author organize the manuscript and provides an easy roadmap for editors, reviewers, and readers to follow while reading the article.

Although, the IMRAD format is widely used, it is not the only format for research articles. For example, the Materials and Methods section can appear at the end of the article in some journals. In others, the Conclusions section is placed at the end. You can find a combined Results and Discussion section, or the Materials and Methods section and the results section can alternate. In some articles, especially in the social sciences, a long Literature Review section may appear near the beginning of the article.

The Abstract is a short powerful summary of your article content. Its function is to showcase your study, highlighting significance of your contribution. The abstract needs to be comprehensive but without giving too much away. It usually explains why you carried out your research, what your aims were, how they were met and what the main findings were.

The abstract is a mini-version of your article. It is a self-contained text, which usually follows the same IMRAD structure. However, depending on the field of study, the abstract can be informative, descriptive, structured or semi-structured. A new trend is getting increasingly popular to present abstracts visually, i.e. graphical and video abstracts.

Task 3. Match the section of a research article (1-5) to its description (A-E).

1	Abstract	A	Describes (if necessary), defends the experimental design, provides enough detail so that a competent researcher can repeat the experiments. Includes the exact technical specifications, quantities and sources or methods of preparation.
2	Introduction	B	Gives some kind of overall description of the experiment providing the big picture. Provides representative data. Constitutes the new knowledge which the author is contributing to the world.
3	Materials & Methods	C	Provides a brief summary of each of the main sections of the paper. States the principal objectives and the scope of the investigation. Describes the methods employed. Summarizes the results. States the principal conclusions.
4	Results	D	Presents the principles, relations and generalizations shown by the research findings. Shows how the results and interpretations agree or contrast with previously published articles. Provides the theoretical implications of the work as well as any possible practical applications. Shows the relationships among the observed facts.
5	Discussion	E	Presents the nature and scope of the problem investigated. Briefly reviews the pertinent literature to orient the reader. States the research methods and, if necessary, the reasons for choice of a particular method. States the principal results of research. States the principal conclusions suggested by the results.

Task 4. Read the list of questions and decide where in a research article you would find the answers. Write one of the following letters in the Section column: I (Introduction), M (Materials and Methods), R (Results), D (Discussion).

№	Questions	Section
1	What objective evidence was obtained from the authors' efforts (observations, measurements, etc.)?	
2	Were the authors able to answer the research question (test the hypothesis posed)? Did the reported observations/ interpretations support or refute observations or interpretations made by other researchers in the field?	
3	How did the result(s) contribute to answering the research question or testing the hypothesis?	
4	What is the authors' key message?	
5	How were the results interpreted? How were they related to the original problem (the authors' view of evidence rather than objective findings)?	
6	What research methods were used to accomplish the main purpose of the article?	

Task 5. Give the English equivalents to the following words and word combinations from the text of Task 2.

1. подробный обзор литературы _____
2. графические и видео _____
3. раздел «Материалы и методы» _____
4. план действий _____
5. автономный текст _____
6. всесторонний _____

Task 6. Fill in the gaps in the sentences with the appropriate English equivalents from Task 5.

1. The IMRaD format helps the author organize the manuscript and provides an easy _____ for editors, reviewers, and readers to follow while reading the article.
2. The _____ can appear at the end of the article in some journals.

3. In some articles, especially in the social sciences, _____ may appear near the beginning of the article.

4. The abstract needs to be _____ but without giving too much away.

5. Abstract is a _____, which usually follows the same IMRaD structure.

6. A new trend is getting increasingly popular to present abstracts visually, i.e. _____ abstracts.

Task 7. Watch the video “The structure of scientific research papers” by Steve Kirk and decide whether the statements below are “True” or “False”.

1. The speaker explains that the structure of scientific papers is known as IMRAD.

2. The introduction section of a research paper is always labeled as "Introduction."

3. The method section describes the subjects and conditions of the experiment.

4. The results section includes figures or tables to illustrate findings.

5. The discussion section is where the paper becomes more specific about the experiment.

6. The abstract is a detailed explanation of the study's methodology.

7. The speaker mentions that the introduction section is useful for learning about the subject.

Task 8. Watch the video “The structure of scientific research papers” by Steve Kirk again and answer the questions below.

1. What is the main purpose of the introduction section in a research paper?

2. How does the method section contribute to the overall understanding of a study?

3. What type of information is typically included in the results section of a research paper?

4. Why is it important for the results section to present numerical facts without interpretation?

5. What is the primary focus of the discussion section in a research paper?

6. Why are references listed at the end of a research paper?

7. How can graphs and figures enhance the presentation of data in a research paper?

Task 9.* Select an article from your target journal and analyze its structure. Complete the table Literature Review Chart with the key sentences from the corresponding sections of the article.

Literature Review Chart

Citation information	
Introduction	
Methods	
Results	
Discussion	

Unit 2. Introduction and Literature Review Sections

Task 1. Answer the questions.

1. What is the purpose of the Introduction Section?
2. How large is the Introduction of a research article?
3. What verb forms are mostly used in the Introduction section?
4. Why do scientists very often write the Introduction Section last?

Vocabulary

rationale – логическое обоснование

scope of the investigation – масштаб (рамки) исследования

paramount importance – первостепенная важность

briefly and precisely – кратко и точно

research design - план исследования

Task 2. Read the text about the Introduction Section and find the answer to question 3 from Task 1.

The Introduction is one of the most important parts of the article. In this part, you state the motivation for your work and give an overall review of your research. If you do it competently, the reader will want to read the whole of your article. The aim of the Introduction is to inform the reader about the rationale behind your work, justifying why your research is an important contribution to the field. You should present the nature and scope of the investigated problem as clear as possible, relating it to the established knowledge. You may want to briefly explain what methods you are using and why. You should also announce the principal results of your investigation and conclusions you make to gain the reader's attention. It is also helpful to orient the reader in the structure of your article.

Your Introduction will be clearer and more logical if you separate your research actions from what the article itself covers. To describe what you have

done, use the following pattern: start with the doer of the action (*we, I, the authors*); use a verb expressing a research action: *measured, computed, etc.* in the past or present tense. To explain the object of the manuscript, use the name of the manuscript (*this paper, this research*) and add a verb in the present tense, expressing a communication action, e.g. *presents, reports on*.

Very often scientists write the Introduction Section last not to overlook some key points. However, it is important to start planning it as early as possible. It is of paramount importance to state the problem briefly, but very distinctly. If you do it briefly and precisely, the reader will be interested in your solution. Your research question or aim should be absolutely clear too, as it directs your research design, guides the choice of literature and stops you from going off in unnecessary directions.

Task 3. Read the text about the Introduction Section again and match the content elements of the Introduction (A-H) to their purposes (1-8).

	<i>Content element</i>		<i>Purpose</i>
A	Provide context for the research area	1	To make the nature and scope of the problem clear; to lead to the need for further investigation
B	Narrow down the context to the investigated problem	2	To orient those readers who are less familiar with your topic and to establish the importance of your work
C	Relate the stated problem to the current state of knowledge / the actual situation	3	To briefly inform the reader about your actions to address the need
D	Present a desired situation / a need to be addressed	4	To state a persuasive motivation for your work; to make it clear what niche you are going to fill in
E	Outline what the authors of the present study have done and how: research actions and methods	5	To announce your principal conclusions which make your contribution significant
F	Emphasize the contribution of the study	6	To prepare the reader for detailed information; to focus reading
G	State the aim of the study	7	To demonstrate that you are familiar with the most important findings in relation to the problem; to give credit to the work of other researchers
H	Describe the structure of the article	8	To clearly show the objective of your research

Task 4. Give the English equivalents to the following words and word combinations from the text of Task 2.

1. общий обзор _____
2. логическое обоснование _____
3. сущность и масштаб _____
4. основные результаты вашего исследования _____
5. исполнитель _____
6. коммуникативное действие _____
7. первостепенное значение _____
8. сделайте это кратко и точно _____

Task 5. Fill in the gaps in the sentences with the appropriate English equivalents from Task 4.

1. In this part, you state the motivation for your work and give _____ of your research work.
2. The aim of the Introduction is to inform the reader about _____ behind your work.
3. You should present _____ of the investigated problem as clear as possible.
4. You should also announce _____ and conclusions you make to get the reader's attention.
5. To describe what you have done, use the following pattern: start with _____ (we, I, the authors).
6. To explain the object of the manuscript, use the name of the manuscript (this paper, this research) and add a verb in the present tense, expressing _____.
7. It is of _____ to state the problem briefly, but very distinctly.
8. If you _____, the reader will be interested in your solution.

Task 6. Watch the video "How to Write the Introduction Paragraph of a research paper" and decide whether the statements below are "True" or "False". Make necessary changes to correct false sentences.

1. The opening paragraph typically occupies two or three paragraphs.
2. The Introduction section must provide necessary background knowledge.
3. After reading your introductory paragraph the reader shouldn't so far be familiar with the main idea of your paper.

4. This section typically contains three parts.
5. You must explain clearly what problem your paper is trying to address or solve.
6. You don't need to establish why the topic is timely or important.
7. At the end of the introductory paragraph you cannot explain the benefits your work will bring to the community.
8. The opening paragraph typically should start with a sentence that piques the interest of your readers.

Task 7. Watch the video “How to Write the Introduction Paragraph of a research paper” again and answer the questions.

1. What is the opening paragraph and what are its main characteristics?
2. What must the introduction section provide?
3. Why is it necessary to narrow down your topic after presenting the general topic of your research?
4. What should you do to define your research problem?
5. How can you establish that your research topic is timely and important?
6. What is the idea of the hook sentence?
7. What are the ways to present familiar information in an interesting manner?
8. Which of the examples presented in the video and script “How to Write the Introduction Paragraph of a Research Paper” appeals to you more? Why?

Task 8. Read the fragment from the video script “How to Write the Literature Review Section of a Research Paper” and answer the questions after the text.

The **Literature Review (LR)** typically appears after the Introduction paragraph of your research paper. Broadly speaking, it is a summary of previous research on your chosen topic. The purpose of the **LR** is to let your audience know what others have done in your line of research. The Literature Review is not about providing a shopping list of past papers. You should structure your **LR** so that it flows logically and provides just the right amount of information to your readers. The length of the **Literature Review Section** varies from discipline to discipline. Typically, in a standard research paper, the **LR** occupies one or two paragraphs. However, if you are dealing with a topic that requires a strong theoretical foundation, then a significantly longer **LR** might be required. Generally, in your **LR**, you can talk about the following things. You can start your **LR** with a broad summary of a previous research on the topic. Then, you can speak about seminal or pioneering works in literature.

Here you can talk about historical papers and famous people who kick started research in your field. Then, you can talk about specific works related to your topic. Here, you can cite prominent papers that are relevant to your specific study. Then you can talk about the most recent papers published in your field. Remember the papers you are claiming to be recent should have been published in the last 2 to 3 years and they should reflect the newest discoveries and theories. Then you can talk about controversial literature related to your topic, if there is any. Here you can talk about studies that led to conflicts and debates among researchers in your field. Towards the end of the **LR**, you have to establish the research gap. A research gap is a problem that has not been addressed or answered in previous studies. You must highlight the drawbacks of previous works and clearly explain what is missing from the literature.

(Some authors distinguish a *Research gap* or a *Problem* section as a separate one).

1. Where in a research paper does **LR** typically appear?
2. What is the purpose of **LR**?
3. How many paragraphs does **LR** occupy in a standard research paper?
4. What can you start your **LR** with according to the text recommendations?
5. When can you cite prominent papers that are relevant to your specific study?
6. What kind of papers can be claimed as recent?
7. What is controversial literature?
8. What must you establish towards the end of the **LR**?

Task 9. Read the example of the Literature Review paragraph and fill in the gaps with phrases from the box below.

<i>proposed an alternative suggestion</i> <i>was further extended</i> <i>in more recent</i> <i>in the first approach</i>
--

“1. _____ proposed by Democritus (400 B.C.), a Greek philosopher, suggested that the matter is made up of small particles called atoms [1]. The method 2. _____ by Dalton (1808) who proposed the atomic theory, known as Dalton’s Atomic Theory [2]. Niels Bohr (1913) 3. _____; that the electrons of an atom revolve

around a centrally placed nucleus [3]. 4. _____ work, physicists have successfully created a “giant atom”, these atoms are created at very cold temperatures and are held together by a weak bond [4].”

Task 10. Match the sentences of the research article Introductions to the structural parts.

1. **Motivation / Relevance** _____
2. **Literature Review** _____
3. **Problem formulation** _____
4. **Solution (Aim of the study)** _____

A. “There exists a vast amount of literature on the topic of consumer behavior [1-5]. Existing literature on the consumer decision-making process generally explores: Personal factors [6-8], Age factors [9-11], Psychological factors [12-15] and situational factors [16-20]. There are very limited published previous research findings related to shopping behaviors of older population. This study, to the best of our knowledge, represents the first attempt to fill the void in the literature”

B. “Extensive academic research addresses the topic of mental health in a wide range of contexts, from medication [1-3] to therapy [4-6]. In recent years, research has only begun to examine the impact of social factors on depression [7-8]. Even more recently, marketing research has begun to examine the role of corporate responsibility in promoting mental health support for employees [9-10]. In fact, in the only two such studies that have been conducted to date on this issue, inconsistent findings have been reported. Obviously, more studies are needed.”

C. “Artificial Intelligence (AI) has received considerable attention in recent years due to its application in various fields such as manufacturing, healthcare, security, etc. In recent years methods have been proposed to produce sufficiently intelligent software capable of reprogramming. Such production presents a number of challenges as AI may threaten human employment in the future. The development of new programmes will facilitate a better understanding of just how far AI can go.”

Task 11. * Select an article from your target journal and analyze its Introduction section structure. Complete the table Literature Review Chart-2 with the key sentences from the corresponding sections of the Introduction.

Literature Review Chart-2

Citation information	
Motivation /Relevance	
Literature Review	
Problem formulation	
Solution (Aim of the study)	

Task 12. Study the table with clichés sometimes used for writing the Introduction Section and create improved alternatives.

Cliché Phrase	Why It's Problematic	Improved Alternative
“In recent years, there has been increasing interest in...”	Vague, non-specific, and overused. Doesn't quantify “interest” or explain <i>why</i> it matters.	
“Many researchers have studied [topic]...”	Fails to engage with the literature meaningfully. “Many” is meaningless.	
“This topic is of great importance.”	Asserts value without justification. Readers need to be convinced, not told.	
“Despite numerous studies, little is known about...”	Often inaccurate (there's usually <i>some</i> knowledge) and dismissive of prior work.	
“The purpose of this study is to...”	Passive, formulaic, and interrupts narrative flow.	
“To the best of our knowledge, no study has...”	Often used defensively; can be easily disproven.	
“This paper aims to fill a gap in the literature.”	Redundant – all good research does this. Saying it doesn't make it true.	
“It is widely accepted that...”	Often false or unsupported. Can misrepresent consensus.	

Cliché Phrase	Why It's Problematic	Improved Alternative
“The literature is vast and complex.”	An excuse for poor synthesis. Avoids the hard work of organizing prior work.	
“In conclusion, this study contributes to the field.”	Too late – this belongs in the <i>conclusion</i> , not the introduction.	

 **Introduction: Why Avoid Clichés?** A strong research article introduction does more than state the topic – it builds a logical, evidence-based case for why the study matters. Clichéd phrases undermine credibility, obscure originality, and signal lazy writing.

Task 13.* Write the Introduction section for the article on your research topic (180-220 words). Cite 5 papers: prominent, the most recent ones and controversial literature related to your topic. Put the numbers of the link to the References in square brackets. Use Appendix for more information.

Unit 3. Materials and Methods Section

Task 1. Answer the questions.

1. What is the purpose of the Materials and Methods Section?
2. What is methodology in terms of research?
3. What should you do to enable other researchers to replicate your research the same way you did it?
4. What two big groups can methods fall into?
5. What does the choice of methodology depend on?

Vocabulary

trustworthiness – достоверность

validity and reliability – достоверность и надежность

quantitative research methodology – методология количественных исследований

qualitative methodology – качественная методология

appropriate – подходящий

Task 2. Read the text about the Materials and Methods Section and say what two big groups methods can fall into.

The purpose of the Materials and Methods section is to provide support for the trustworthiness and significance of claims of your research article. The methodology – the main principles behind the set of methods used in the research – describes the techniques and experimental procedures used to identify, select, process, and analyze relevant information about the research topic. In a research article, the Materials and Methods section allows the reader to make an objective judgement on the study's overall validity and reliability.

The Materials and Methods section typically includes information about participants, materials, study design and procedure. To enable other researchers to replicate your research the same way you did it, you should:

- Describe the subjects/materials used in the study

- Explain how the materials were prepared for the study or how the subjects were selected
- Describe the research protocol
- Explain how measurements were made and what calculations were performed
- State what statistical analysis was done to analyze the data.

“Materials” refer to what was examined (*e.g. humans, animals, plants, tissue preparations*) and also to various treatments (*e.g. drugs, gases*) and instruments (*e.g. chemicals, testing instruments, technical equipment*).

The description of preparations, measurements and the protocol should be organized chronologically.

Methods can be defined as techniques of collecting and analyzing data. Methods can fall into two big groups: quantitative and qualitative. The quantitative research methodology is based on numbers and mathematical calculations. The qualitative methodology entails gathering data and drawing conclusions from interpretations and perceptions. The choice of methodology depends on the research question or hypothesis. Methodology is part of your study design which should be clearly described. This part of the article should include the protocol for collecting data, i.e. how the experimental or study procedures were carried out, and how the data were analyzed. You should describe your methods and procedures of measurements in detail so that the reader can easily follow your procedures and clearly understand why a particular method is the most appropriate for your study. If you choose a traditional or widely used method, you certainly do not need to state the obvious. For less traditional approaches, reasoning for their choice is required.

Task 3. Match possible subheadings of the Materials & Methods Section (1-10) to the questions they may answer (A-J).

1	Overview of the research	A	How did I analyze numerical data to maximize, interpret and use the results obtained?
2	Theoretical justification	B	What materials and equipment did I use (modifications) and where did this equipment come from (vendor source)?
3	Population	C	How did I apply the chosen method(s) to gather and analyze the data?
4	Location	D	What was my research limited by?

5	Restrictions / Limiting conditions	E	Where did I carry out this research and what characteristics did this location have?
6	Sampling technique(s)	F	What did I study?
7	Procedures	G	What was the general purpose of my research? How can I summarize my research?
8	Materials	H	What factor or condition was varied or changed throughout my research?
9	Statistical analysis	I	What references to the literature could I give not to describe something in detail?
10	Variables	J	How did I design sampling?

Task 4. Give the English equivalents to the following words and word combinations from the text of Task 2.

1. достоверность и значимость утверждений _____
2. необходимая информация _____
3. достоверность и надежность _____
4. чтобы другие исследователи могли повторить ваши исследования _____
5. методология количественных исследований _____
6. качественная методология _____
7. определенный метод является наиболее подходящим для вашего исследования _____

Task 5. Fill the gaps in the sentences with the appropriate English equivalents from Task 4.

1. _____ the same way you did it, you should describe the subjects/materials used in the study.
2. _____ is based on numbers and mathematical calculations.
3. The purpose of the Materials and Methods section is to provide support for the _____ of your research article.
4. _____ entails gathering data and drawing conclusions from interpretations and perceptions.
5. In a research article, the Materials and Methods section allows the reader to make an objective judgement on the study's overall _____.

6. The reader can easily follow your procedures and clearly understand

7. Experimental procedures are used to identify, select, process, and analyze _____ about the research topic.

Task 6. Fill the gaps in the dialogue with the collocations from the list and be ready to reproduce it (*structure, detailed equations, measurements, to bear in, endeavor, a new method, Materials and Methods Section (M & M-s), replicate, the make, reproduce, reference, are extending*).

How to Start Writing a Research Article?

A: Hello! Could you give me a piece of advice how to start writing a research article? It looks a rather challenging 1) _____ to me so far.

B: Hello! Certainly, I'll be glad to share my experience on doing it. I believe, if you are going to write a research article or paper, it is a good idea to start writing it with the 2) _____. This is generally the easiest section to write because you are simply reproducing what you did in your experiments.

A: It sounds good! What are the most important things 3) _____ in mind while writing M & M-s Section?

B: This Section should have enough details so that other researchers in your field can 4) _____ your experiments and 5) _____ your results.

A: I see. What can you say about the 6) _____ of the M & M-s Sections?

B: If you are writing an Engineering Sciences research paper, in which you are

7) _____, then you have to detail the entire procedure step by step. However, if you 8) _____ previous method, then you can 9) _____ the paper and just explain what changes you have made.

A: Actually, I'm going to write a paper that deals with 10) _____.

What kind of information is typically included in the M & M-s Section in this case?

B: You can try to answer these questions: What equipment was used to perform the measurements? What was 11) _____ and the model of the equipment? How many technicians took measurements? How experienced were the technicians? Then you can talk about the actual measurement procedure.

A: What is the most important issue while writing the M & M-s Section?

B: Most importantly, you must list all the calculations in the form of 12) _____ and formulas so, that readers know exactly how the data was produced.

A: Thank very much for the instructions. Now I know what to start with in writing my research paper.

B: Any time! Let me know if you have any other questions or doubts.

Task 7*. Role-Play. Work in pairs, make a dialogue on the topic “Experiment” according to the information of the role description presented below:

Student 1: You are preparing to conduct an experiment to verify data obtained through computer modeling. Answer your research advisor’s/colleague’s questions regarding:

- a) the purpose of the experiment;
- b) the materials and equipment used in the experiment;
- c) the methods and the reasons for choosing them;
- d) the practical/theoretical significance of the results obtained;
- e) the conditions and possible difficulties;
- f) the reasons for any discrepancies between the expected and actual results.

creative software

Student 2: You are the research advisor or a colleague of the person you are speaking with. Ask about:

- a) the purpose of the experiment;
- b) the materials and equipment used in the experiment;
- c) the methods and the reasons for choosing them;
- d) the practical or theoretical significance of the results obtained;
- e) possible difficulties;
- f) reasons for any discrepancy between expected and actual results.

Task 8. Watch the video “How to write the Materials and Methods Section”, or read the script to the video and answer the questions.

1. Why is this section the easiest for writing?
2. What is one of the most important things to bear in mind while writing the M & M-s Section?
3. Why is it necessary to provide all details in a logical order?
4. What kind of methods can be presented in the M & M-s Section?

5. What information will you include writing a paper that deals with measurements?
6. What questions should you answer writing about measurement procedures?
7. In what form must you list all the calculations? Why?
8. What information would you typically include in the M & M-s Section writing a survey paper?
9. What are characteristics of a survey?
10. How can be questionnaire administered?

Task 9. Fill the gaps in the sentences from the video “How to write the Materials and Methods Section”, using word combinations from the box.

<i>detailed equations and formulas</i>	<i>top-level summary of the method</i>
<i>the questionnaire</i>	<i>the questionnaire</i>
<i>have enough detail</i>	<i>pilot or preliminary studies</i>

1. One of the most important things to bear in mind while writing the Materials and Methods section is that it should _____ so that other researchers in your field can replicate your experiments and reproduce your results.

2. If you are writing an Engineering Sciences, research paper in which you are introducing a new method, your Materials and Methods section will include the following information: You can start with the _____.

3. You can also include information about any _____ you conducted before the full study.

4. Most importantly, you must list all the calculations in the form of _____ so, that readers know exactly how the data was produced.

5. Then you can talk about how _____ was administered. If it is an online survey, how did you get _____ to the participants?

Task 10. Read the example of the Materials and Methods paragraph and fill in the gaps with phrases from the box below:

<i>by means of</i>	<i>due to</i>	<i>by using</i>	<i>to fix</i>	<i>without</i>
--------------------	---------------	-----------------	---------------	----------------

Probabilistic Latent Semantic Analysis (PLSA) is an approach that has been released after LSA method 1) _____ some disadvantages that have been found in LSA. Jan Puzicha and Thomas Hofmann introduced it in the year 1999. PLSA is a method that can automate document 2) _____ indexing which is based on a statistical latent class model for factor analysis of count data, and also this method tries to improve the Latent Semantic Analysis (LSA) in a probabilistic sense 3) _____ a generative model. The main goal of PLSA is to identify and distinguish between different contexts of word usage 4) _____ recourse to a dictionary or thesaurus. It includes two important implications: First one, it allows to disambiguate polysemy, i.e., words with multiple meanings. Second thing, it discloses typical similarities 5) _____ grouping together words that shared a common context.

(Alghamdi R., Alfalqi K., A Survey of Topic Modeling in Text Mining: International Journal of Advanced Computer Science and Applications, Vol. 6, No. 1, 2025)

Task 11. Study the table with clichés for writing the section “Materials and Methods” and write improved alternatives.

Materials and Methods Section: Clarity Over Ceremony

Cliché Phrase	Why It’s Problematic	Improved Alternative
“The experiments were carried out as follows.”	Empty filler. Adds no information; sounds like a textbook.	
“Standard protocols were followed.”	Vague and unverifiable. What’s “standard”?	
“All reagents were purchased from commercial sources.”	Redundant – assume this unless specified otherwise.	
“Samples were analyzed in triplicate.”	Too vague – <i>how</i> ? What was measured?	
“The apparatus was calibrated before use.”	Assumed. Doesn’t add value unless <i>how</i> or <i>with what</i> is stated.	

 Golden Rule for Methods:

“Could another researcher replicate this exactly from your description?”

If not – rewrite.

Task 12*. Write the Materials and Methods Section for your research article (150 – 200 words).

Unit 4. Results Section

Task 1. Answer the questions.

1. What are the two main components of the Results section?
2. What type of logic should the text of the Results section follow?
3. Why are the results, presented in a research article, called the heart of the article?
4. What is critical for clear and precise writing about research results?

Vocabulary

a straightforward description – четкое описание

without bias – непредвзято

without cluttering up the text – не загромождая текст

in an appropriate manner – надлежащим образом

clear and precise writing – четкое и точное письмо

completed research – завершённое исследование

Task 2. Read the text about the Results Section and choose an appropriate subheading (A-F) to each paragraph from the list below.

1. _____

The Results section is a straightforward description of exactly what you observed or discovered during your research. Typically, there are two main components of this section. First, you should provide the overall description of your findings arranged in a logical sequence without bias or interpretation, and, second, you should present your data in tables and/ or figures (charts, graphs, maps, images, etc.). The text of the Results section should follow the logic of your research and provide answers to your research question(s). Important negative results should also be reported.

2. _____

The results presented in a research article are the heart of the article. If the results are not convincing, original or useful, there will be no point in

writing an article to share them. However, the way the results are presented must be factual and informative.

3.

The Results section should be organized around a series of graphical elements sequenced to present your key findings in a logical order. Tables and figures are effective ways of conveying much information without cluttering up the text. Guides for Authors vary greatly when it comes to presenting data. In some cases, separate sections containing detailed instructions about exactly how to lay out tables and figures will be provided, whereas in others, authors will simply be advised to format tables and figures in an appropriate manner.

4.

Use appropriate for your data figure type. **Pie charts** are used to highlight proportions of a total or whole. **Column** and **bar charts** are used to compare the values of different categories when they are independent of each other. **Line charts** are used to display a sequence of variables in time or space or to display other dependent relationships. **Radar charts** are used when categories are not directly comparable.

5.

Your key results might include trends, differences, similarities, correlations, maximums, etc. You may need to compare the values, qualify the changes and analyze your observations. Therefore, knowing how to make comparisons is critical for clear and precise writing about research results.

6.

As the Results section describes completed research, the Past Simple tense (active or passive) is the natural choice. However, when you refer to tables and figures, you should use the Present Simple Tense, since your graphical elements continue to exist in your article.

A. Making comparisons

B. Laying out tables and figures

C. Two main components of the section

D. The Present Simple Tense versus the Past Simple Tense

E. Appropriate figure type

F. Convincing Results

Task 3. Choose the most appropriate word in brackets to fill in the gaps.

1. The _____ (*low, lower, the lowest*) the frequency, the _____ (*much, more, the most*) the relaxation processes are completed, giving rise to total damping.

2. We now know that there are different types of responses to core – holes, where inter-molecular or intra-atomic screening constitutes a _____ (**more appropriate, the most appropriate, quite appropriate**) terminology.

3. Our results showed (Figure S2) that among all the tested polymer coatings, CMD-coated nanoparticles had _____ (**little, less, the least**) binding to cells.

4. _____ (**Farther, Further**) analysis by transmission electron microscopy (TEM) revealed _____ (**-----, more, the most**) complex than cubic shape of the particles, similar to previously reported hematite nanocrystal.

5. Figure 3B shows that when placed in the media with a similar pH and phosphate content (PBS or the culture medium), the release of surface – absorbed Dau proceeds _____ (**more faster, much faster, more fast**), apparently, due to the phosphate-induced degradation of the upper layer of nMOFs (Fig.3B, solid red line).

6. The results of MTT cell viability test in the presence of various concentrations of Dau and Dox showed (see Figure S5) that the indicated parameters for the BT-474 and MCF-7 lines were _____ (**close, more close, the closest**). In addition, because of the _____ (markedly great, markedly **greater, markedly greatest**) cytostatic activity of daunorubicin, it was used for _____ (**farther, further**) experiments.

7. Since surface atoms have _____ (**more, -----, three time**) fewer neighbors than bulk atoms, hybridization should act to make the core-level Bes of surface atoms _____ (**nearly, half as much, -----**) smaller than those of bulk atoms.

8. The Al₆(Mn, Cr) particles are approximately _____ (**two times, two time, just**) smaller than Al₆Mn particles.

Task 4. Fill in the gaps with the correct form of the verb in brackets paying attention to the voice (Active or Passive):

1. Several identical Fe-27Ga samples (1) _____ (**anneal**) during 40 min at 1000°C, i.e. in the A2 range of the equilibrium Fe – Ga phase diagram. Then the samples (2) _____ (**water quench**) to prevent diffusion – controlled phase transitions during cooling at lower temperatures. In contrast with the quenched samples, one sample (3) _____ (**furnace cool**) to room temperature.

2. Figure 2 (4) _____ (**highlight**) the typical stress – strain curves for Al-BNNPs composites and the relationship between the room temperature tensile strength, yield stress values, and BNNPs concentration in composites. The maximum tensile strength of 150 MPa (5) _____ (**observe**) for Al-based composite materials with 4,5 wt% of BNNPs.

3. The dendritic microstructure of the as-cast alloy (6) _____ (**completely / break down**) and the Si particles (7) _____ (**distribute**) homogeneously throughout the Al matrix in both FSP1 and FSP2. The particles also (8) _____ (**grow**) in size and (9) _____ (**become**) equiaxed. Both the FSP conditions also (10) _____ (**eliminate**) casting porosities.

4. Figure 3 (11) _____ (**present**) the S–N curves generated from the fatigue tests for the 3 microstructural conditions – Cast, FSP-AGG, and FSP-fine. The comparison (12) _____ (**show**) at three temperatures – room temperature (RT) in Figure 3(a), 150°C in Figure 3(b), and 200°C in Figure 3(c).

5. Two in-situ STEM aging experiments (13) _____ (**carry out**) to investigate the evolution of primary and secondary precipitates during thermal exposure. The first experiment (14) _____ (**perform**) with a lamella taken from a coarse-grain region, with no primary Al₃Zr phase, in an as-fabricated Al-3.60Mg-1.18Zr (wt%, Batch 2) sample. The sample (15) _____ (**ramp**) from room temperature to 400°C in 160s and then (16) _____ (**hold**) at this temperature for 480s. Figure 6 (17) _____ (**show**) a STEM-EDIX map taken after quenching the sample back to room temperature after 480s at 400°C.

Task 5. Read the extract from the video script “How to Write the Results Section of a Research Paper?” and find the English equivalents of the Russian words and collocations after the text.

Trends and patterns describe the general change in some variable in relation to another variable. Finally, you can talk about any reanalysis of data you had to perform in order to reconfirm your findings. Let’s look at some examples. Here is an example from an engineering research paper where the authors talk about preprocessing: *“The following is a brief summary of the preprocessing steps applied prior to analysis. The data were screened for outliers and such data points were set to missing. Subsequently, the missing data were filled by linear interpolation. The data were transformed using the Box method and then normalized to zero mean and unit standard deviation. The smoothing was applied at the final step to remove the noise in the data.”*

The authors are saying that they performed a series of steps before conducting the actual analysis. They are saying that they filled in the missing data using linear interpolation and then they transformed and normalized the data. Finally, they applied data smoothing to remove any noise in the data. As you can see the authors are very transparent and are detailing everything they did to the data before the actual analysis.

1. переменная _____
2. повторный анализ данных _____
3. предварительная обработка _____
4. отклонения _____
5. недостающие данные _____
6. блок-метод _____
7. линейная интерполяция _____
8. выравнивание данных _____
9. честный _____
10. фактический анализ _____

Task 6. Fill the gaps with the appropriate English equivalents from Task 5.

1. Trends and patterns describe the general change in some variable in relation to another _____.
2. The data were screened for _____.
3. Finally, they applied _____ to remove any noise in the data.
4. They filled in the missing data using _____.
5. The data were transformed using the _____.
6. The authors performed a series of steps before conducting the _____.

Task 7. Watch the video “How to Write the Results Section of a Research Paper?” and answer the questions below.

1. Why is the Results section relatively easy to write?
2. What can you start this section with?
3. What types of statistics must you try to use to present your data?
4. How is it possible to summarize large amounts of data in a visually pleasing way?
5. What do trends and patterns describe?
6. What can you do to reconfirm your findings?
7. What must you clearly explain while presenting your findings?

8. When should researchers present the data in figures tables instead of putting it directly in the text?

9. How to choose between a figure or a table?

Task 8. Match the types of visuals (1 – 6) with illustrations (a – f).

1. a histogram

3. a line graph

5. a scatter plot

2. a bar chart

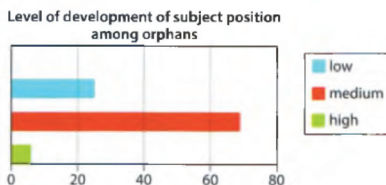
4. a table

6. a pie chart

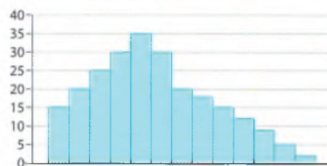
a

Tasks completed	Before experiment (%)	After experiment (%)
Bachelor's students	13.6	43.4
Master's students	15.7	87.6

b

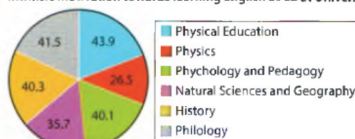


c

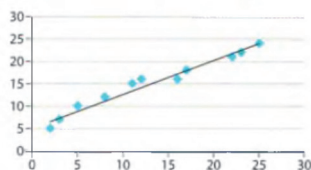


d

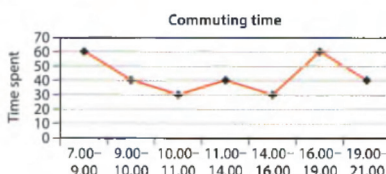
Intrinsic motivation towards learning English as L2 at University



e



f



Task 9. Match the types of visuals (a – f) with the purpose they are used (1 – 5) in academic texts.

a) a histogram

d) a table

b) a bar chart

e) a scatter plot

c) a line graph

f) a pie chart

1. A _____ is used to show exact numbers.

2. A _____ is used to display relations between items.

3. A _____ or _____ are used to show trends.
 4. A _____ is used to show proportions of a whole.
 5. A _____ is used to investigate the possible relationship between two variables that both relate to the same “event”.

Task 10. Read the description of the histogram below and match the sentences (1-6) to the questions they answer (a-f).

- a) refers to specific information in the diagram? ____
 b) adds some details? ____
 c) summarizes the main findings? ____
 d) makes the reference to the histogram? ____
 e) makes a prediction? ____
 f) introduces the main findings? ____

1) The chart in Figure 1 shows the number of women researchers in five OECD countries. 2) The vertical Y-axis illustrates a percentage of female researches; the horizontal X-axis presents a time period from 2004 to 2010. 3) It can be clearly seen that the figures for all countries has remained stable with some fluctuations between approximately 11% in Korea in 2004 and 42% in the Slovak Republic in 2010. 4) It also demonstrates that in two of the countries, the number of female researches has not changed. 5) In general, countries experienced no dramatic changes in the rates of women researches. 6) If the trend continues, we may expect an overall increase in the number of women involved in research.

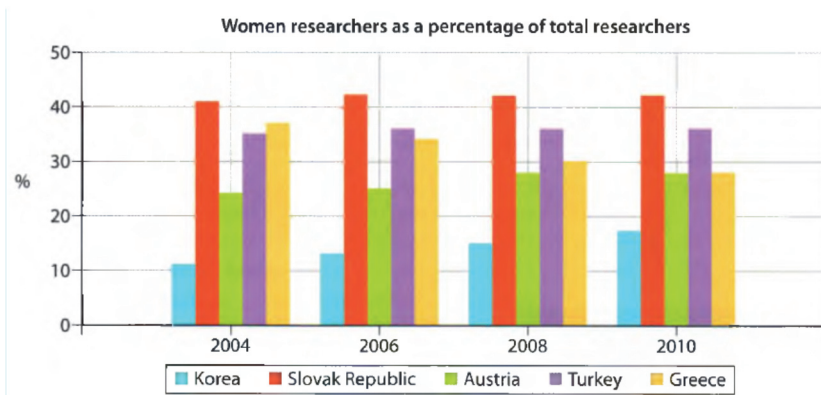


Fig. 1. Women researchers as a percentage of total researchers

Task 11. Read the phrases below and say what they are used for in a description of visual information.

- As shown in (Figure 1, Table 2, etc.) ...
- The diagram outlines ...
- The figure above/ below illustrates...
- The pie chart represents...
- The line graph depicts/ indicates...

Task 12. Read the description of the line graph of articles submitted to international journals (Figure 2) and fill in gaps with phrases from the box below:

articles	grows	Figure 2	2012
2007	between	rose	to
fall	stable	slightly	

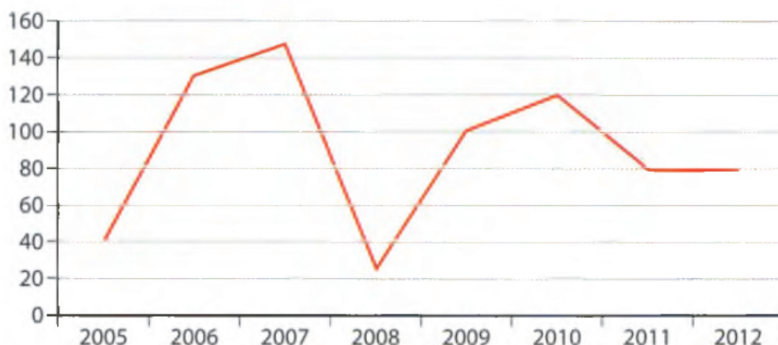


Fig. 2. Articles submitted to international journals

The line graph in 1) _____ shows the number of 2) _____ submitted to different international journals by researchers from our institution 3) _____ 2005 and 4) _____. First, the number of articles 5) _____ sharply from 2005 6) _____ 2006. Then, as you can see, it increased 7) _____ between 2006 and 2007. Clearly, the number of articles reached its peak in 8) _____. There was a sharp 9) _____ in 2008. After this, we experienced gradual

10) _____. From the middle of 2011, the number of articles remained 11) _____.

Task 13.* Write a description of a graph, table or a diagram showing trends or changes on your research topic. Use about 100 – 120 words. You can use the Useful language below.

Language Useful: describing trends

There was a (very)	slight gradual steady considerable sharp dramatic rapid	increase growth rise decrease decline drop fall	<i>with a definite period of time from (July) to (September) during (March) between (2021) and (2026)</i>
	noticeable considerable	fluctuation	
Results, prices, numbers, etc.	increased grew rose decreased declined dropped fell	insignificantly slightly gradually steadily sharply dramatically rapidly	
	fluctuated	considerably	

Unit 5. Discussion and Conclusion Sections

Task 1. Answer the questions.

1. What does the Discussion section aim at?
2. What section of a research article does the Discussion section echo with? Why?
3. What does the Discussion section start with?
4. What is the aim of the Conclusion section?
5. How many paragraphs are usually there in the Conclusion section?
6. What does the Conclusion section not contain? Why?

Vocabulary

standalone – обособленный, автономный

insight – видение, проникновение в сущность

overlap – пересечение, совпадение

concise – краткий, лаконичный

to the point – по существу, по делу

Task 2. Read the text about the Discussion and Conclusion Sections and answer the questions from Task 1.

In different disciplines, the Results, Discussion and Conclusion sections can be standalone, or there may be different combinations of them, e.g. *Results and Conclusion*, *Discussion and Conclusion* or *Results and Discussion*. It is important to remember that no matter how the article is structured, each of these sections has a definite purpose. The Results section contains your answer to the research question(s). The Discussion section aims at demonstrating how your key findings fit in to what is already known. The Conclusion section describes new insights or value of your research contribution. Structuring your article, you need to avoid unnecessary overlaps.

The Discussion section is one of the most important parts of your article. It aims to interpret your findings in light of what is already known in the field

about the research topic. It explains new insights which emerged as a result of your study. This section echoes with the issues raised in the introduction. You are supposed not only to restate the hypothesis or research question but clearly explain what your study adds to the understanding of the research problem you reviewed in the introduction. That is the place where your opinion enters the overall picture of prior research. The section usually starts with specific research findings and moves to some general research implications. The Discussion section may comprise such content elements as critical assessment of your major results, comparison with other studies, your key message and directions for future research.

The aim of *the Conclusion section* is to help the reader understand the value of your research. It contains not only the summary of the major findings or a restatement of the research problem. You should also synthesize and highlight the key points, note down theoretical or practical applications and indicate/ recommend areas for further research. Depending on the logic of the narrative, implications for further research can be given in the Discussion section. Decide what section suits better for this purpose. The Conclusion section should be concise and to the point. Usually, one to three well – developed paragraphs are sufficient. Just as the Introduction provides the first impression, the Conclusion is to make a lasting impression on the reader. While writing a Conclusion section, think about what you want the reader to remember about your research. Present only key findings and specific implications and/ or applications which emerged from results. Make sure that the Conclusion section does not contain any new arguments or data, and not end quotation. Make your Conclusion strong so that your original contribution to the field can potentially be cited.

Task 3. Find English equivalents to the following words and word combinations in the text from Task 2.

1. направлен на _____
2. согласовываться с _____
3. в свете чего-либо _____
4. перекликаться с _____
5. применение _____
6. для дальнейшего исследования _____
7. кратко и по делу _____
8. продолжительное впечатление _____
9. быть процитированным _____

Task 4. Fill the gaps in the sentences with the appropriate English equivalents from Task 3.

1. The section usually starts with specific research findings and moves to some general research _____.
2. The Conclusion section should be _____.
3. Make your Conclusion strong so that your original contribution to the field can potentially _____.
4. It aims to interpret your findings _____ what is already known...
5. The Discussion section _____ demonstrating how your key findings _____ what is already known.
6. This section _____ the issues raised in the introduction.
7. Depending on the logic of the narrative, implications _____ can be given in the Discussion section.
8. Just as the Introduction provides the first impression; the Conclusion is to make _____ on the reader.

Task 5. Read the examples of the Discussion section from research articles and retell one of them in the 3rd person plural. Watch the video and compare your version with the authentic one.

1. *"Our aims were to compare the effectiveness and user-friendliness of different storm detection software for storm tracking. On the basis of these aims, we ran multiple experiments with the same conditions using different storm detection software. Our results showed that in both speed and accuracy data, 'software A' performed better than 'software B'".*
2. *"We found the largest differences in ocean heat from 1993 to 2021 averaged about 0.16 Watts per Square Meter. The differences in ocean heat in these years can, in part, be attributed to excess heat from the earth's surface being passed back to the ocean. This could also be due to the rising amounts of greenhouse gases preventing heat radiated from the earth's surface from freely escaping into space."*
3. *"First and foremost, results indicate that there is a strong relationship between early childhood education and future career success. The findings of this study have clear applications for policymakers and educators. This research adds to existing literature that supports the idea that investing in early education can have long-term benefits for individuals and society as a whole."*

4. *“The results contribute insights with regard to the management of wildfire events using artificial intelligence. One could easily argue that the obvious practical implication of this study is that it proposes utilizing cloud-based machine vision to detect wild fires in real time, even before the first responders receive emergency calls”.*

Task 6. Match the examples of the Discussion section from research articles (1-4) from Task 5 to the components of the Discussion section (1-6). Read and analyze the script of the video “How to write the Discussion Section of a research paper” and find two more examples.

1. research summary
2. interpretation of findings
3. comparison with literature
4. implications of your work
5. limitations
6. future work

Task 7. Watch the video “How to Write the Conclusion Section of a Research Paper” and name the components the section typically contains.

1. _____
2. _____
3. _____
4. _____
5. _____

Task 8. Analyze the examples from the scientific articles, presented in the video script and find the examples of the Conclusion Section components from Task 7.

1. *“In summary, the use of natural light in office buildings is investigated in this study. The results show that there is a strong correlation between the use of natural light and the productivity of employees. The results demonstrated in this work provide a new perspective on the importance of natural light in office buildings and have important implications for the design and construction of future office buildings.”*

2. *“In summary, this study provided evidence for the impact of climate change on sea level. Specifically, research showed that the sea level has risen*

by an average of 8 inches over the last century, and the rate of sea level rise has accelerated in recent decades. To our knowledge, this is one of the most comprehensive studies to date assessing the impact of climate change on sea level. Therefore, these findings have important implications for coastal communities and infrastructure planning.”

Task 9. Study the cliché phrases presented in the table below, sometimes used for writing the Discussion Section. Find improved alternatives for them.

Discussion Section: Interpret, Don’t Inflate

Cliché Phrase	Why It’s Problematic	Improved Alternative
“Our findings are novel and groundbreaking.”	Overblown. “Novel” must be proven by comparison to literature.	
“This study provides new insights into...”	Vague. What <i>insight</i> ?	
“These results are in agreement with previous studies.”	Too easy. <i>Which</i> studies? How?	
“This could have important implications for...”	Speculative and grandiose.	
“Further research is needed.”	Lazy, overused, and unhelpful.	

 **Golden Rule for Discussion:** “*Connect your results to the literature. Explain discrepancies. Propose mechanisms. Avoid hype.*”

Task 10. Study the cliché phrases sometimes used for writing the Conclusion Section. Find improved alternatives for them.

Conclusion Section: End with Impact, Not Repetition

Cliché Phrase	Why It's Problematic	Improved Alternative
“In conclusion, we have shown that...”	Redundant – you already showed it in Results/Discussion.	
“This work contributes to the field of [topic].”	Vague and empty. What <i>exactly</i> did it contribute?	
“More studies should be conducted.”	Same as “further research is needed.”	
“We hope this study will inspire future research.”	Emotional, unprofessional.	
“The limitations of this study are...”	Often buried at the end, sounding defensive.	

💡 **Golden Rule for Conclusion:** *“Answer: So what? What’s the takeaway? What now?”* Don’t restate. Don’t apologize. Don’t flatter. *Conclude with purpose.*

Task 11*. Write the Discussion and Conclusion Section for your research article (150 – 200 words). Remember about structural parts of the Section.

Unit 6. Title, Key Words and an Abstract

Task 1. Answer the questions.

1. Why are titles and key words of research articles so important?
2. Why does an abstract play a pivotal role in communicating research?
3. What is the function of an abstract?
4. What does an abstract need to be like?
5. What does an abstract explain?
6. What is the structure of an abstract?

Vocabulary

searchable – востребованный

citable – цитируемый

pivotal role – ключевая роль

to screen manuscripts – отбирать рукописи

at the outset – в самом начале

a self – contained text – самодостаточный текст

Task 2. Read the text about the Title, Key Words and an Abstract and answer the questions from Task 1.

Key words point researchers to relevant articles. Titles come first when people search for an article by typing suitable keywords on the topic. Having read the title, readers decide on the relevance of your article to their research. The better and clearer the title is formulated, the more readers will go through to the **Abstract**, and hopefully, to the whole article. That is why, it is necessary to think carefully about your title as it makes your research searchable and potentially citable.

If you succeeded in capturing readers' attention with your title, they would get through to your abstract. For the reader, the abstract is a "bridge" to your article. Some readers never cross it. The abstract plays a pivotal role in communicating research. Many journal editors screen manuscripts only based on abstracts. If the abstract does not grab their attention or does not

make a good first impression, editors are likely to reject the article at the outset. The **Abstract** is a short powerful summary of your article content. Its function is to showcase your study, highlighting the significance of your contribution. The **Abstract** needs to be comprehensive but without giving out too much away. It usually explains why you carried out your research, what your aims were, how they were met and what the main findings were.

The abstract is a mini – version of your article. It is a self – contained text which usually follows the same IMRaD structure. However, depending on the field of study, the abstract can be informative, descriptive, structured or semi – structured. A new trend is getting increasingly popular to present abstracts visually, i.e. graphical and video abstracts. To write an effective abstract, you will have to go back to the draft of your article to make sure that the information presented in the abstract is consistent with the content of the article. The abstract should show the best qualities of your research. It is advisable to make your abstract concise but informatively rich. You should also strictly follow the journal requirements for the abstract (word limit, type of abstract, subheadings, if recommended, etc.).

Task 3. Find English equivalents to the following words and word combinations in the text from Task 2.

1. играет ключевую роль _____
2. востребованный _____
3. самодостаточный текст _____
4. отбирать рукописи _____
5. в самом начале _____
6. привлечение внимания читателей _____
7. цитируемый _____
8. соответствующие статьи _____
9. представить результаты вашего исследования _____
10. ключевые слова по теме _____

Task 4. Fill the gaps in the sentences with the appropriate English equivalents from Task 3.

1. The abstract _____ in communicating research.
2. It is necessary to think carefully about your title as it makes your research _____ and potentially _____.
3. Key words point researchers to _____.

4. Titles come first when people search for an article by typing suitable

_____.

5. If you succeeded in _____ with your title, they would get through to your abstract.

6. Many journal editors _____ only based on abstracts.

7. The abstract is _____ which usually follows the same IMRaD structure.

8. If the abstract does not grab their attention or does not make a good first impression, editors are likely to reject the article _____.

9. The Abstract function is _____, highlighting the significance of your contribution.

Task 5. Watch the video “How to Write the Abstract Section of a Research Paper?” and answer the questions.

1. What are three reasons why you should spend a decent amount of time writing your abstract?

2. What questions should the abstract clearly answer?

3. What are two types of abstracts? What are their characteristic features?

Task 6. Read the dialogue on writing an abstract, fill in the gaps with the words and word combinations from the list (*self-sufficient, a summary, the abstract, available for free, passage, a research paper, be available, a rough idea, reviewers, submitting, to purchase, structured, implications, unstructured*).

A: Hello! How are you? Have you already submitted your abstract to the Students' Conference Proceedings?

B: Hi! I'm quite well, thank you. What for to send an abstract of

1) _____ so early? The conference will be held in April and it is winter yet. I have plenty of time to write a small **2)** _____ on my research topic.

A: **3)** _____ is one of the most important sections of your research paper. People normally see your abstract first because it will **4)** _____ in conference programs and various databases. This section should be the last thing you write because this is **5)** _____ of entire paper along with your final conclusions.

B: Do you mean, I will be able to write a good abstract just after having written the whole research paper?

A: Definitely. The abstract is like a mini paper. The abstract should be 6) _____, and just by reading the abstract, the reader should get 7) _____ of what you have done in the paper. There are three reasons why you should spend a decent amount of time writing your abstract.

B: Let me guess what these reasons are. After 8) _____ my paper to a journal, the editor reads the abstract and decides whether to send my paper for review or to reject it. The 9) _____ read the abstract first, if it is badly written, then they won't bother wasting time reading the rest of the paper. Am I right?

A: Absolutely! What is more, the abstracts are 10) _____. Readers will read the abstract first and then decide whether 11) _____ the full version of the paper or not. The abstract should clearly answer the following questions. 1) Why are you writing this paper? 2) What did you do? 3) What are your results? 4) What are the 12) _____ of your findings?

B: It sounds persuasive. I really shouldn't put off my research article writing any more. I wonder if I should write a 13) _____ or unstructured abstract?

A: You are right, there are two types of abstracts: structured abstract and unstructured abstract. Most journals follow the 14) _____ format. This is basically writing your abstract as a continuous paragraph. According to the requirements of the upcoming Students' Conference, everyone, who wants his/her paper to be published, should structure the abstract under four headings:

1) Background and objectives, 2) Methods, 3) Results, 4) Conclusions.

B: Thank you for the sharing the information. Obviously, it's high time I started writing my research paper. When my article is written, I will be able to create a structured, self-sufficient abstract. Have a good day! See you!

A: Have an inspiring, successful writing! Talk to you later. Bye!

Task 7. Match the headings of the structured abstract (1-4) to the paragraphs (A-D). Watch the video or read the video script «How to Write the Abstract Section of a Research Paper?» and check yourself.

1. Background and objective

2. Methods

3. Results

4. Conclusions

A. Mean BMI decreased in the vitamin D group (22.3 ± 0.2) compared to the placebo group (26.02 ± 0.3), and a significant effect was observed ($p \leq 0.001$).

B. In a randomized placebo-controlled trial, we examined the effect of vitamin D supplements on weight loss over 24 weeks in 100 obese women with BMI as a primary outcome.

C. The findings have proven that vitamin D supplementation can be used as a potential therapeutic option for weight loss programs.

D. Obesity and vitamin D deficiency are two conditions that usually occur simultaneously in patients. However, it is unclear, whether vitamin D supplements aid in weight loss among obese patients.

Task 8. Read the example of the unstructured abstract and find the sentences referring to the components: Background and objective, Methods, Results, Conclusions

1. The quality of video is affected by errors occurring at various stages of the production chain and are very challenging to fix. 2. Automatic error correction methods based on the principles of motion flow estimation and image correlation are proposed. 3. The new algorithms outperform existing techniques with superior performance in quality and speed. 4. The new technique eliminates the visual inspection process and automatically restores the video integrity.

Task 9. Select an article from your target journal and analyze an abstract according to the following points:

1. The title of the research article, the author(s), journal, date of the research paper publication, the abstract comes from.

2. Whether the abstract is structured or unstructured.

3. How the author(s) answer(s) the questions about

1) the reason of writing the paper;

2) what investigation they performed or experiments they did;

3) how they present the results of the research;

4) what implementations of the findings are.

Task 10. Read a sample scientific article on computer science and replace the highlighted clichés in the text with improved alternatives:

- ✗ “Novel” → ☑ *“We introduce X, which improves Y by Z% over prior work”*
- ✗ “Significant leap” → ☑ *“Our method reduces latency by 22% compared to baseline”*
- ✗ “We believe” → ☑ *“The results suggest...”* or *“Data show...”*
- ✗ “More research is needed” → ☑ *“Future work should evaluate X under condition Y”*
- ✗ “Standard protocols” → ☑ *“We used mAP@0.5 (COCO eval protocol), FPS measured over 100 runs”*

“A Novel Deep Learning Framework for Real-Time Object Detection on Mobile Devices”

Introduction

In recent years, there has been a surge of interest in deploying deep learning models on edge devices due to the increasing demand for real-time applications such as autonomous driving, surveillance, and augmented reality. Object detection is a fundamental task in computer vision, and **many researchers have attempted to solve it** efficiently. Despite the proliferation of deep learning techniques, **there remains a critical need for** lightweight models that can run efficiently on resource-constrained mobile platforms. **This paper presents a novel** deep learning framework called FastDetectNet, which aims to address this challenge. **Our approach is simple yet effective,** and **we believe it will set a new standard** for mobile object detection. **We hope this work inspires future research** in efficient AI deployment.

Materials and Methods

We used a dataset consisting of 10,000 images from COCO and Pascal VOC, which are **widely used in the community**. The model was implemented in PyTorch and trained on an NVIDIA RTX 3090 GPU. We **followed standard training procedures**: Adam optimizer, learning rate of 0.001, batch size of 16, and trained for 100 epochs. For inference, we tested on three mobile devices: iPhone 13, Samsung Galaxy S21, and Google Pixel 6.

We compared our method with YOLOv5, SSD, and EfficientDet. All metrics were computed using **standard evaluation protocols**. We **ran each experiment three times and took the average** to ensure reliability.

Results

As shown in Table 1, FastDetectNet outperforms all other methods in both accuracy and speed. It achieves a mean Average Precision (mAP) of 78.4%, which is higher than YOLOv5 (75.1%), SSD (71.3%), and EfficientDet (74.8%). More importantly, it runs at 45 FPS on the iPhone 13, which is **much faster** than the others. The results are very impressive and **clearly show that our model is superior**. We also noticed that the model works better on daytime images – probably because of lighting conditions. Figure 2 shows a **nice visualization** of detection results.

Discussion

Our proposed FastDetectNet is **groundbreaking** and represents a significant leap forward in mobile AI. **The results are consistent with prior work**, confirming that model pruning and quantization improve efficiency. **Our method is better because** we used a new attention mechanism – although we didn't ablate it to prove it. The main limitation is that we only tested on three devices. **More research is needed** to evaluate performance on older phones or under low-light conditions. **Future work could explore** integration with edge TPUs or federated learning. This work **opens up exciting new possibilities** for real-time AI on smartphones.

Conclusion

In conclusion, we have presented FastDetectNet, a **novel and efficient** deep learning model for real-time object detection on mobile devices. Our experiments demonstrate superior performance compared to existing methods. This work **contributes meaningfully to the field** of edge AI. We believe our framework **will be widely adopted** by developers and researchers alike. **Again, more research is needed** to extend this work to other domains.

Task 11* Write the Abstract Section of the research article on the topic of your investigation.

Task 12* Write a research article on the topic of your investigation.

Supplementary Vocabulary and Grammar Exercises

Verb Tenses and Their Meaning in Academic Writing

Task 1. Look through the text first. Then fill in the gaps with the forms given in the list: *is proposed, have co-evolved, proposes, is based, is expected, is, has made, is based, pose, can be managed, review, can deliver, infer*

Artificial intelligence (AI) (1) _____ remarkable progress in the last several years. Computers can now see, listen, and drive a car, in some cases as well as humans, or even better. Machine translation (2) _____ no longer an embarrassment. Much of today's AI (3) _____ on deep learning (DL), a brain-inspired machine learning framework (Goodfellow et al., 2016, Sejnowski, 2018). Together with other brain-inspired and statistical approaches, AI (4) _____ to further benefit human life and society in areas such as information services, manufacturing, mobility, environment, healthcare, and science. However, rapid advances also (5) _____ large risks. Here we (6) _____ expected benefits and potential risks of AI and neurotechnologies and how these technologies (7) _____ so as to minimize undesirable outcomes.

The article (8) _____ on presentation and discussions at the International Symposium on AI and Brain Science held online in October 2020. In the following sections, we first (9) _____ how AI and brain science (10) _____ and what else can be learned from the brain to guide future progress of AI. WE then review how progress in AI and neurotechnologies (11) _____ helpful innovations, with a focus on the potential of AI to promote scientific discoveries. We then consider the risks associated with AI and neurotechnologies, including international misuse, unintended side effects, and loss of control.

We then consider ethics and governance of AI, with two specific viewpoints. For the issue of goal misspecification by AI systems, a theoretical framework of assistance games (12) _____, in which AI agents continually (13) _____ human preferences, rather than pursuing a specific goal. Another view (14) _____, by learning from the history of human society, to implement democratic peer reviewing among open-source, explainable AI agents, to avoid catastrophes by over-concentration of power.

Task 2. Put the following collocations into the table. (A collocation is a particular combination of words frequently used together, for example: *adjective + noun, verb + preposition + noun, noun + noun*)

Baseline survey, have the potential, controlled trials, scale up, data loss, further exploded, rigorous trials, validity checks, experience suggests, data accuracy, large-scale survey, extensive way, appeal for, conduct experiments, negligent conduct, conform to, easily altered, language background, rapid progress, theories evolve, question exists, loosely based, challenge the statement

adjective + noun	
verb + noun	
noun + noun	
verb + preposition	
adverb + adjective	
noun + verb	

Task 3. Using the online Synonyms Dictionary at <https://www.synonyms.com> , find more formal equivalents to the words in bold.

informal	formal
do a survey	
start research	
get results	
give an example	
find out the reasons	
a lot of evidence	
good effect	
bad impact	
large amount	
maybe	

Various Expressions for Cause – Effect

1. Affect (v) = to influence or cause smb. or smth. to change, is used without a preposition:

e.g. Smoking affects your health.

Effect (n) = the result of an influence, e.g., to have an effect on smth.:

Smoking has a bad effect on your health.

2. Cause (v) = to make smth happen, especially smth. bad, e.g., to cause problem, an illness, harm, damage, etc. **To cause** is frequently used in its passive form – **to be caused by smth.:**

Greenhouse effect is caused by the presence of certain gases in the air.

3. Because is used at the beginning of a clause, when giving a reason for smth.:

I was late because the traffic was bad.

In academic English, **because** is rarely used at the beginning of a sentence.

Because is used before a noun or a noun phrase:

I was late because of the traffic.

4. Due to = because of, it is mostly used in formal or official contexts:

All flights have been cancelled due to weather conditions.

5. To result from smth. = to be caused by it.

The rise in population has resulted from a higher birth rate.

To result in something = to make smth. happen, e.g.,

A higher birth rate has resulted in the rise in population.

As a result is used as a linking device and surrounded by commas:

As a result/ as a consequence of = because of:

Profits have declined as a result of the recent drop in sales.

Task 4. Read the excerpts and find words or phrases which help to express cause – effect relations. The first two are already underlined.

1. Historical evidence shows that market shares of (energy) technologies follow logistics growth patterns. The growth in early phases is slow due to a long list of interconnected problems.

2. These different energy types may affect the drive for renewables as they may or may not 1) result in energy import dependency; 2) lead to concern over environmental impact; and/or 3) make it easier or more difficult to integrate substantial shares of renewables into power mix.

3. Market size and growth variables had significant effects on PV developments.

4. This is not a credible method, as we do not know their ultimate market shares.

5. For our dataset, which includes a wider variety of countries, we did not manage to find a sufficiently exhaustive set of domestic energy prices for all major fuel types and years, and we therefore decided to exclude these.

6. Lacking data quality may be as much of a reason for the presence or absence of significance as the existence of an actual relationship between the energy price and outcome variables.

7. Because of different data availability, we combined data from Eurostat, UN data, and the BP review.

8. Domestic sector growth may lead to positive feedback loops, as the technology itself matures, but also because manufacturing and maintenance industries grow in size and become more competitive.

Task 5. Choose the appropriate connector out of the three given. You do not need to know all the words in the sentences to make the correct choice.

1. Here, we showed that removal of Foxg1 in adulthood **resulted** / **led** / **affected** in decreased dendritic complexity.

2. The formula cannot be used directly for E calculation because **of** / **because** / **cause** the fluorescence level of the starting material F0 is unknown.

3. The ionospheric plasma convection is generally **led** / **resulted** / **caused** by the Formula omitted drift.

4. In such cases aid may limit the strains on the domestic tax base, and thus prevent costly distortions. **Therefore** / **Due to** / **Because**, aid has the potential to address the technology and non-technology aspects of TFP.

5. NAICS 447 illustrates good tracking to official estimates in three regions, but not in the South. **As a result** / **Because** / **Because of**, caution is recommended in interpreting estimates relying on regional data that include the South.

6. Most households (77 percent) indicated that access to land was most important for housing, and 43 percent stated housing as the main reason **of** / **for** / **in** moving.

7. Our results suggest that some aid flows, particularly technical cooperation, have a significant and positive effect **to** / **in** / **on** the productivity in the region.

8. The outcome **results** / **affects** / **leads** to exactly the same amount of deadweight loss given in the equation.

9. In contrast, **due** / **because** / **result** to the seasonal transition from the boreal summer to the winter monsoon, SON rainfall contributes up to 60-80% of the annual precipitation in the VCC.

Task 6. Read the excerpts from scientific articles and underline the relative clause in each one. Then underline the noun which the relative clause refers to.

1. As part of an attempt to bridge this gap, this study is driven by the motive to understand how pedagogical innovations such as task-based language instruction can be influenced by teacher beliefs. Drawing on the perspective of research on teachers' thinking, which aims to inform language teaching pedagogy and teacher education, the study employed multiple data sources (focus group, lesson plan, and stimulated recall interview) to tap into the connection of six Vietnamese EFL instructors regarding language output and interaction.

2. The second part is mainly concerned with formalisms within which it can be proved that a strategy will achieve a goal. The concepts of situation, fluent, future operator, action, strategy, result of a strategy and knowledge are formalized.

3. This study reports on beliefs of four Japanese teachers of English, who engaged in teaching-oriented postgraduate study in the UK, and focuses on their expectations about international programs.

4. This study addresses mature teachers as students abroad and focuses on the challenges to their existing professional beliefs triggered by their foreign academic sojourn. The participants are Japanese teachers of English who attended MA programmes in the UK.

5. Theoretically, knowledge about second language acquisition (SLA) principles is argued to offer teachers a tool for trial in their classrooms, and the opportunity to reflect and change. However, research that focuses on the effect of SLA theory training has revealed little impact on teacher beliefs about language learning and teaching.

6. They should include the names of two persons to whom references may be made.

7. The people who this report is addressed to will have to consider its proposals carefully.

8. We define effective professional development as structured professional learning that results in changes in teacher practices and improvements in student learning outcomes.

9. The respondents that were asked about the MBA programme helped to choose the best courses.

Task 7. Match the sentences halves. Underline the word a relative clause refers to and the pronoun. One pronoun is wrong. What is it? Why is it wrong?

- | | |
|---|--|
| 1. The goal of the inventions
For institutions is to develop
staff training strategies | A. which also raised £2000
for the Appeal. |
| 2. Thus, for instance,
If a business has a particular
Customer with | B. who have been unsuccessful in
achieving all the learning outcomes
must have the opportunity to be
re-assessed after appropriate
consolidation |
| 3. The committee headed by
Dame Gillian Brown
held a Literary Lunch
in honor of Penelope Fitzgerald
a fascinating occasion, | C. that will effectively prepare people
for burnout avoidance and thus
reduce the institutional costs of
burnout effects. |
| 4. Fourthly, pupils | D. that will do a lot of business, it
might be appropriate to negotiate
special terms for all contrasts
between the business and that
customer. |

Video Scripts

Unit 1 The structure of scientific research papers

Okay, the purpose of this video is to teach you about the basic structure of scientific and medical research papers. The structures of these kinds of papers are so common, that it has the name IMRaD as you can see here and you can see there's even in a Wikipedia page about it. IMRaD stands for "Introduction, Methods, Results and Discussion."

Research papers have four main sections as it is shown in this diagram from Wikipedia. The sections are again: Introduction, Methods, Results and Discussion, although, sometimes the names might be a little different. Sometimes, the "Introduction" doesn't have a name in the paper, it's just the beginning; and the "Methods" section is sometimes called "Materials and Methods". And the "Discussion" section might be a "Discussion" only and a "Conclusion" after that, or it might be "Results and Discussion" together and called "Results and Discussion". But basically, these four parts are always there.

Now you can also see in the shape of this diagram that it is wider in the top and narrower in the middle. The meaning of that is that at the beginning of the Introduction, the paper is more general, and then it becomes more specific to the topic of your study. And then in the "Discussion" section it goes back from specifically about your experiment to going more general again. Here is an example of one research paper about contagious yawning in chimpanzees.

If I highlight the different areas, then you can see that there are four main sections: Here is the **Introduction**, here is the **Materials and Methods**, here is the **Results** and here is the **Discussion** in green. Before the Introduction up here, you see that we have the **Abstract**. The Abstract is one paragraph that explains the main point of the study. And before that we have of course, the **Title** and the names of the authors. Here we have three authors: Anderson, Myowa Yamakushe and Matsazama.

Next, I want to explain the main point of each of these sections. The “Introduction”: this part of the paper answers the question “Why was this study done?” and inside the Introduction it will 1) introduce the main topic, 2) give a summary of previous research, 3) formulate a clear problem to be solved, 4) explain the hypothesis, 4) explain the main purpose of the study. This section is also useful to learn about the subject if you don’t know much about it. Because it should teach you the important background information and review the most important previous research about the topic. The end of the Introduction has the main purpose of the study. That’s important to know if you are looking for that point.

The next section is the “Methods” and this section answers the question: “How was the study done?” It explains 1) the subjects, 2) the conditions and treatment, 3) the procedure to do the experiment and what was measured. This section is usually a kind of a boring one, because it’s a very detailed. But readers want to be able to check how someone did the experiment so, that they know that they can trust the results. And they might have different interpretation of the results. So, when you first read a paper you might skip this section if you just want to know the results and you don’t have any particular questions that you want to answer.

The next section is the “Results”. This section answers the question: “What did the study find?” So, it explains: 1) the results, 2) any problems with data collection, 3) any other interesting trends with the data, 4) mentions figures or tables that show the results and, in the sentences, it will mention those figures or tables. An important point to know about the Results is that this section explains the results without any interpretation, just the numerical facts and the statistical results. So, that readers can think for themselves about the meaning. The graphs, as you can see here, are usually part of the “Results” section, and notice that the graphs have a clear explanation called a caption. The purpose of this is that you can read the caption and look at the graphs and figures, the graph or the picture and you can understand it without having to read anything else.

The last section is the “Discussion” and it answers the question: “What is the meaning of the results?”: 1) it reviews the main results of the study; 2) compares the results with previous research; 3) discusses possible reasons and interpretations of the results; 4) gives suggestions for further research, applications and conclusion. This is usually the most interesting section because it tells you the main points and what they mean. But to be able to trust these conclusions you’re gonna need to read the rest of the paper to find out more of the details of how they did the study.

Last, there is after the discussion the last part of the paper – it's the list of references and there are all of the references that are mentioned in the paper so you might see like right here it says, you know, at this point here reported by Anderson and Meno in 2003 and you can look here and see. By the way, Anderson Junior Anderson is also the author of this paper. So, this is his own previous research. That's the main part of this video, but I also just want to quickly show some other examples so, you can see what other papers look like. ...

Unit 2. How to Write the Introduction Paragraph of a Research Paper?

Hi, this is Chelsea from Ref-n-Write. In this video we will learn how to write the opening paragraph of your research paper. The opening paragraph is the first paragraph of your research paper. The opening paragraph is where you want: 1) to attract your readers and draw them into your paper, 2) this typically occupies one or two paragraphs, 3) the introduction section must provide necessary background knowledge, 4) so that the readers can understand the motivation behind your experiments and appreciate your findings. After reading your **introductory paragraph**, the reader should be familiar with the main idea of your paper and understand why your topic is important. This section typically contains four parts: 1) You can start with a hook. Hook is nothing but an attractive opening statement. 2) Then, you have to provide a broad introduction to your topic. Here, you can define the general topic of your research 3) Then the next step is to narrow down your topic and reveal the specific focus of your paper. 4) Then you have to define your problem. You must explain clearly what problem your paper is trying to address or solve. 5) Then you need to establish why the topic is timely or important. Here you can explain the benefits your work will bring to the community and also talk about how it will advance the field and the current understanding of the topic.

Let's spend more time talking about hooks. The opening paragraph typically should start with a sentence that piques the interest of your readers. This is called a hook. The idea is to hook your readers and reel them in just like using hooks with bait to capture a fish in a pond. These are statements designed to grab the reader's attention and make them want to read more of your paper. One thing to bear in mind is that most people reading your paper will have some background knowledge about the topic. So, you have to find a way to say things that people already know in a slightly different and interesting manner. The trick is to use statistics, facts, quotations and questions to make

things interesting to the readers. These are called hooks. Now let's look at some examples of hooks. Here is an opening statement. Here we are trying to establish that breast cancer is a pretty bad disease. This statement is not very punchy, it is pretty bland and doesn't convey the seriousness of the situation. Now let's use some statistics and rewrite the statement. Look at this statement now: **"In 2020 43,700 females died from breast cancer in the US. It is the most common cancer in women accounting for 30% (or 1 in 3) of all new female cancers"**. Now, the readers will instantly understand how deadly breast cancer is because you are using numbers to put things into perspective. This will get readers attention straight away. And they will be keen to read more of your paper to find out if you are proposing something to fix this terrible situation.

Here is another example. Here is an opening statement about global warming. There is nothing new here. Most people know the global warming is bad. Now, let's frame this as a question with some facts and numbers. **"Did you know that the last decade was the hottest in 125,000 years and there is more CO2 in our atmosphere than at any time in human history?"** How about this? This sounds way better because it conveys the urgency of the situation. There are so many different ways to write hooks for your paper. It takes a bit of practice to acquire this skill. The best place to start will be to go through the opening statements of highly cited papers to get an idea. Let's look at some examples of introduction paragraphs. The examples we have chosen are quite short, so that it's easy for you to understand. We have chosen examples from different fields and of different formats.

Here is an example: ***"Artificial Intelligence (AI) has received considerable attention in recent years due to its application in various fields such as manufacturing, healthcare, security, etc. In recent years methods have been proposed to produce sufficiently intelligent software capable of reprogramming. Such production presents a number of challenges as AI may threaten human employment in the future. The development of new programmes will facilitate a better understanding of just how far AI can go."***

This is a passage from a social sciences research paper and it is about AI which stands for Artificial Intelligence. We are providing a broad overview of the topic in the first couple of lines. We are talking about the popularity of the AI and the recent advancement that have happened in the field. Then we move on to the problem definition. The problem is about AI threatening the human employment in the future by taking up all the jobs. Then in the next statement, we are talking about the importance of exploring the issue so that we can understand how far AI will impact our lives in the future. You can see

that this passage ticks all the boxes. There is relevant background information, a clear definition of the problem, and information about the importance of conducting research on this topic.

Look at this example: ***“The Asian market is growing and relatively unexplored segment in the marketing literature. Approximately 60.5 million Asians lived in the United Kingdom in 2022, and this will grow at rate of approximately 13.19% over the next five years [1]. The purchasing power of this demographic is expected to reach over \$2.26 trillion by 2025 [2]. Researchers and practitioners should therefore be interested in the increasing importance of this market and the potential it has for businesses.”*** This passage is from a marketing and business paper. In this example we are using a lot of numbers to establish why it is important and timely to conduct research in this domain. We are telling the readers that there are millions of Asians living in UK and their numbers are increasing every year. They have this huge spending power and not many researchers have looked into how and where they spend their money. So, the time has come to explore the market. A lot of interesting data and facts are provided to the readers. And that is the reason why this is such a good introduction.

Here is example from a medicine paper: ***“Obesity is a world-wide disease. In 2020 more than 2 billion adults, 18 years and older, were overweight. There is a general consensus in the research community that there is a strong association between obesity and vitamin D. This represents an important and timely topic because obesity is currently fifth greater risk of mortality, and vitamin D deficiency has been associated with variety of diseases. Better understanding of this link will greatly aid medical practitioners in effective treatment of obese patients.”*** We are starting the passage with a hook by providing an interesting statistic about obesity. After starting with a broad statement, we are narrowing down the topic. We are dropping a hint that our paper is to do with vitamin D. And obesity with the next statement, we are establishing the importance of the topic. We are saying that a lot of people are dying due to obesity and vitamin D is causing a lot of health issues, so we must do something about it. Then in the final statement of the intro paragraph we explain how conducting research in this field will benefit the community. In our case, doctors will be able to prescribe better treatment options for obese patients. This is a well written introduction paragraph that nicely introduces the main idea of your paper and tells the readers why your topic is important.

Unit 2. How to Write the Literature Review Section of a Research Paper.

Hi, I am Chelsea from Ref-n-Write. In this video we will learn how to write the **Literature Review Section** of a Research Paper. The **Literature Review (LR)** typically appears after the Introduction paragraph of your research paper. Broadly speaking, it is a summary of previous research on your chosen topic. The purpose of the **Literature Review** is to let your audience know what others have done in your line of research. The Literature Review is not about providing a shopping list of past papers. You should structure your **Literature Review** so that it flows logically and provides just the right amount of information to your readers. The length of the **Literature Review** Section varies from discipline to discipline. Typically, in a standard research paper, the **Literature Review** occupies one or two paragraphs. However, if you are dealing with a topic that requires a strong theoretical foundation, then a significantly longer **LR** might be required. Generally, in your **LR**, you can talk about the following things. You can start your LR with a broad summary of a previous research on the topic. Then, you can speak about seminal or pioneering works in literature. Here you can talk about historical papers and famous people who kick started research in your field. Then, you can talk about specific works related to your topic. Here, you can cite prominent papers that are relevant to your specific study. Then you can talk about the most recent papers published in your field. Remember the papers you are claiming to be recent should have been published in the last 2 to 3 years and they should reflect the newest discoveries and theories. Then you can talk about controversial literature related to your topic, if there is any. Here you can talk about studies that led to conflicts and debates among researchers in your field. Towards the end of the LR, you have to establish the research gap. A research gap is a problem that has not been addressed or answered in previous studies. You must highlight the drawbacks of previous works and clearly explain what is missing from the literature.

Let's look at some examples of LR. Typically, in a LR you start with broad issues and then you narrow it down to specific issues. The you keep narrowing it down until you reach the articles that are most relevant to your research question. Here is an example of a standard LR. *“Extensive academic research addresses the topic of mental health in a wide range of contexts, from medication [1-3] to therapy [4-6]. In recent years, research has only begun to examine the impact of social factors on depression [7-8]. Even more recently, marketing research has begun to examine the role of corporate responsibility in promoting mental health support for employees [9-10].*

In fact, in the only two such studies that have been conducted to date on this issue, inconsistent findings have been reported. Obviously, more studies are needed. This example is from a social sciences research paper. The authors have started broadly by saying that research on mental health is quite varied and it ranges from medication to therapy. Then they narrowed down the topic to the impact of social factors on depression and cite a few papers. Then they further narrow it down to the role of corporate in providing mental health support for their employees. Now, this directly relates to their research problem. Then they explain the research gap. They are saying that there aren't many previous studies on the topic. In fact, there are only two studies that have been conducted on this topic so far, the results of those studies are inconsistent. As you can see, the review flows very well. It goes from broad to specific and nicely finishes off with the research gap. One of the important qualities of a good literature review is sequentially explaining the events that led up to a current situation.

Here is an example from a Physics research paper about atomic theory. In this example the authors have summarized almost 1000 years of research work in 3 to 4 lines. ***“In the first approach proposed by Democritus (400 B.C.), a Greek philosopher, suggested that the matter is made up of small particles called atoms [1]. The method was further extended by Dalton (1808) who proposed the atomic theory, known as Dalton’s Atomic Theory [2]. Niels Bohr (1913) proposed an alternative suggestion; that the electrons of an atom revolve around a centrally placed nucleus [3]. In more recent work, physicists have successfully created a “giant atom”, these atoms are created at very cold temperatures and are held together by a weak bond [4].”*** The authors have chronologically provided major theories presented by prominent authors over time. They start with the first approach presented by Democritus in 400 B.C. Then they talk about Dalton’s and Bohr’s theories published in the 18th and 19 centuries respectively. Finally, they talk about the most recent work in the field. Pay attention to the use of phrases such as ***“in the first approach”, “was further extended by”, “proposed an alternative suggestion”, “in recent work”***. These phrases are used to connect the events together and make the text flow better. This passage is a very good example that demonstrates how to logically structure your LR.

When you are dealing with large amounts of literature, it’s a good idea to group them by ideas before presenting them in your LR. Here is an example from a business paper that deals with the consumer behavior of the older population. ***“There exists a vast amount of literature on the topic of consumer behavior [1-5]. Existing literature on the consumer decision-making***

process generally explores: Personal factors [6-8], Age factors [9-11], Psychological factors [12-15] and situational factors [16-20]. There are very limited published previous research findings related to shopping behaviors of older population. This study, to the best of our knowledge, represents the first attempt to fill the void in the literature”. The author starts with the broad statement saying that there is a lot of literature on the topic of consumer behavior. Then the author has grouped articles by ideas and then presented them in the text. As you can see that almost twenty references are cited in the first two lines. Then the author establishes the research gap and explains what is unique about their study. They are saying that there are plenty of studies about general consumer behavior. However, there are very limited studies that specifically deal with the consumer behavior of the older population, and their study represents the first attempt to investigate this issue. You can follow this format of **LR** when you are short of space, but have to include a lot of references.

Here is an example of **LR** from a medical paper: *“Most studies examining the mechanisms linking cycling to male infertility have had conflicting results and is a heavily debated topic [1-2]. Until last decade, there was a general consensus among researchers that cycling improves male fertility [3-9]. The results of the recent large-scale multicenter study by Elan et al [10], in contrast to previous studies, showed cycling weakening the prostate tissue causing infertility among males. Due to lack of clear evidence, the link between cycling and male infertility remains unclear.”* Here the authors talk about a controversial and debated topic in literature. The authors say that the evidence linking cycling and male fertility is somewhat mixed. Some studies say that cycling improves fertility since it's a good exercise, whereas other studies say that it is bad because it actually damages the prostate. The authors say that there is a lack of consensus among researchers about this and we are still not sure which claim is correct. So, there is a research gap here. As you can see, the authors have made their case for further research very elegantly by providing evidence that supports both sides of the argument.

Unit 3. How to Write the Materials and Methods Section of a Research Paper?

Hi, this is Chelsea from Ref-n-Write. In this video we will see how to write the Materials and Methods section of your research paper. This is generally the easiest section to write. Because you are simply reproducing what you did in your experiments. It doesn't require a lot of imagination from your

part. It is always a good idea to start writing your research paper with the Materials and Methods section. One of the most important things to bear in mind while writing the Materials and Methods section is that it should have enough detail so that other researchers in your field can replicate your experiments and reproduce your results.

You should provide all the steps in a logical order. So that your readers can follow your description easily. The structure of the Methods section will very much depend on your discipline. If you are not sure about the structure, then the best place to start will be to go through the Methods section of some previous published papers from your chosen journal. We will look at some examples of Materials and Methods structure in different disciplines. If you are writing an Engineering Sciences, research paper in which you are introducing a new method, your Materials and Methods section will include the following information: You can start with the top-level summary of the method. You can try to answer these questions: Are you proposing a new method or are you using a standard method from the literature? Or are you extending a previously published method? If so, is it your previous work or work published by a different author? Then you can talk about the reasons for choosing the method.

You can quote previous papers that have used this method successfully to support your arguments. Then you can talk about the actual implementation details of the methods. The amount of information you provide will depend on the type of method you are proposing. If you are proposing a new method, then you have to detail the entire procedure step by step. However, if you are extending a previous method, then you can reference the paper and just explain what changes you have made to the previous method in order to adapt it to your work. Then you can talk about how the methods were validated to confirm that they are suitable for your research.

You can also include information about any pilot or preliminary studies you conducted before the full study. Then you can explain how you propose to test and evaluate the methods to prove that they are better than the existing methods. Here you can talk about metrics and statistical tests you will be using to evaluate your method. If you are writing a paper that deals with measurements, you would typically include the following information in your Materials and Methods section. You can start by talking about the experimental setup. You can try to answer these questions: What equipment was used to perform the measurements? What was the make and the model of the equipment? How many technicians took the measurements? How experienced were the technicians?

Then you can talk about the parameters that were measured during the experiment. Then you can talk about the actual measurement procedure. How were the samples prepared for the measurements? How many measurements were taken?" Were the measurements repeated for consistency? Was there a time interval between successive measurements? Then you can talk about measurement conditions and constraints. Were the measurements performed in room temperature or under special conditions? Were there any practical difficulties while performing the measurements? If so, how did you overcome it? Most importantly, you must list all the calculations in the form of detailed equations and formulas so, that readers know exactly how the data was produced.

If you are writing a survey paper, you would typically include the following information in your Materials and Methods section. You can start by talking about your participants. Who is your target population? What are their demographics? How did you recruit them? How did participants provide consent for your study? What sampling method did you use to select the participants? Then you can talk about the survey type. Was it a phone interview? Was it a personal interview? Was it an online survey or was it a written survey? Then you can talk about the questionnaire design. How did you choose the questions? How many questions were there? What type of questions were they? Were they open ended questions or closed ended questions or rating scale questions or a mixture of different types of questions? Then you can talk about how the questionnaire was administered. If it is an online survey, how did you get the questionnaire to the participants? Did you email them or did you post the survey forms? If you are doing a personal interview, how did you conduct the interviews? Was it 1 to 1 interview or was it done in batches, or did you use focus groups? How did the participants behave during the interview? Then you can talk about the questionnaire testing. Did you test your questionnaire before the main study? Did you have to make any changes after initial testing? Did you have to translate the questionnaire into multiple languages? And then finally, you can talk about different types of statistical tests you used to analyze the survey responses.

Unit 4. How to Write the Results Section of a Research Paper?

Hi, this is Chelsea from Ref-n-Write. In this video we will learn how to write the Results section of your research paper. This is a relatively easy section to write. You simply have to organize your results and write them up in a logical sequence. This section typically contains the following components. You can start this section with some information about data preprocessing.

You can also talk about any difficulties you encountered while collecting or processing the data. Then you can start presenting your main findings in a logical sequence. When reporting your results, you must try to use descriptive statistics and inferential statistics to present your data. For example, mean, standard deviation, confidence intervals, p value, etc. You must try to present your results in figures and tables wherever possible. This is a fantastic way to summarize large amounts of data in a visually pleasing way. You must highlight any important trends and patterns in the data.

Trends and patterns describe the general change in some variable in relation to another variable. Finally, you can talk about any reanalysis of data you had to perform in order to reconfirm your findings. Let's look at some examples. Here is an example from an engineering research paper where the authors talk about preprocessing: ***“The following is a brief summary of the preprocessing steps applied prior to analysis. The data were screened for outliers and such data points were set to missing. Subsequently, the missing data were filled by linear interpolation. The data were transformed using the Box method and then normalized to zero mean and unit standard deviation. The smoothing was applied at the final step to remove the noise in the data.”***

The authors are saying that they performed a series of steps before conducting the actual analysis. They are saying that they filled in the missing data using linear interpolation and then they transformed and normalized the data. Finally, they applied data smoothing to remove any noise in the data. As you can see the authors are very transparent and are detailing everything they did to the data before the actual analysis.

Let's look at some examples of how to present your main findings. While presenting your findings, you must clearly explain (1) what did you find? (2) what did you not find? (3) What did you find that you were not expecting? This is an example from a psychology research paper: ***“The results indicate significant differences between classical music and pop music in terms of their effects on memory recall and cognition ($p < 0.05$). However, there is no significant difference across the groups for the emotional response to the music ($p > 0.05$). It was surprising to find that both types of music elicited similar levels of fatigue in both groups ($p > 0.05$).”***

In this paper the authors are investigating the effects of different types of music on people. The authors say that they found significant differences between classical and pop music in terms of memory recall. And then they are saying that they didn't find any differences in terms of emotional response. Finally, they are saying that they were quite surprised to find that both types

of music fatigued the listeners at the same rate. It is very clear from their tone that the last result was a bit unexpected. You can see that the authors have presented their findings in an unbiased manner without any interpretation. They have listed the positive, the negative and the neutral results logically in the text. This is how it should be done.

Now let's look at another example. This snippet is from a social sciences research paper investigating the relationship between social media and mental health: ***“The results of this study indicate significant differences in anxiety levels between high social media usage and moderate social media usage ($p < 0.05$). The average time spent by high social media users (5.2 ± 2.2) was considerably higher than that of moderate social media users (3.2 ± 2.2). The odds of sleep disturbance were significantly greater for high social media users (odds ratio, 2.12; 95% CI, 1.81-2.17) compared to moderate social media users (odds ratio, 1.14; 95% CI, 1.07-1.18).”*** I want you to pay specific attention to the descriptive and inferential statistics used throughout the text. The authors say that there is a significant difference in anxiety levels between high social media users and moderate social media users. Since the authors have used the word ***significant***, they have specified the ***p*** value of the statistical test they used to ascertain this. Then they are providing the actual values of the amount of time spent by both groups on social media. They have presented the data in the form mean plus or minus standard deviation, which is the standard specific way to represent this type of data. In the final statement, they talk about the odds of both groups experiencing sleep disturbances. They have provided the odds ratio along with the confidence intervals. The use of relevant statistics is very important while writing your Results section. It offers two benefits. Number one, it will help you summarize the data in a meaningful way. And number two, it will make your text sound more credible.

Here is an example, where authors are reporting the results of a questionnaire study related to people shopping behavior: ***“The majority of respondents over 70% of the total, preferred shopping online. The second-ranked option was ‘supermarket’ (20%). The lowest-ranked was ‘wholesale shops’ (5%). However, less than half (48.6%) believed that online products are inferior to supermarkets. The average response was ‘4: Fairly Confident’ for the overall rating of confidence in shopping online.”*** The authors have presented the results in a variety of ways. They have used the percentage response to present the main findings. Pay attention to how the authors are very selective in terms of what they have presented in the text. They are just letting the readers know: “What were the first, second and last ranked options?”

Rather than talking about all the options. Because this is likely to be more interesting to the readers. If they want more information, the readers can always refer to the table that contains the full data. The authors have used average response instead of percentage response to present the next set of results. They are saying that on average the participants were fairly confident about shopping online. I want you to pay specific attention to the use of phrases like *the majority of respondents, less than half, the average response was*. The authors are using these phrases to simplify the narrative for the readers rather than just providing the numbers. Your goal while writing this section should be to present your results as clearly as possible to your readers.

One of the important components of the Results section is figures and tables. Try to present your data in figures and tables wherever necessary. One of the common questions asked by students is when should they present the data in figures and tables instead of putting it directly in the text? The best way to answer the question is: “If you cannot present your data and your text in one or two lines, then you should consider putting it in a figure or a table. Another common question that comes up is how to choose between a figure and a table for presenting the data. The best way to answer this question is: *“If you want to illustrate the trends and patterns in the data, then a figure is the best option. If you want to show the actual values or present a lot of numerical information in your paper, then a table might be the best way to go”*. Here is an example. Look at the way the authors have used tables and figures to describe their results. In the first couple of lines they are talking about a trend in their data that relates to the change in temperature over time. They are constantly referring to the figure to get their points across to the readers. And finally, in the last sentence, they are telling the readers that the actual numerical data is provided in the table and they can refer to it if they want. This is a standard way to use figures and tables in your research paper: *“The figure 1.1, the values are plotted as a function of time. The two peaks in the plot correspond to the maximum and minimum temperature values. The specific values obtained for each experiment are given in table 2.”*

Unit 5. How to Write the Discussion and Conclusion Sections of a Research Paper?

In this video we will see how to write **the Discussion Section** of your research paper. This is one of the most important sections of your research paper. There is where you 1) interpret your results, 2) highlight your contributions, and 3) explain the value of your work to your readers. This section

typically contains the following parts: research summary, interpretation of findings, comparison with literature, implications of your work, limitations, future work. It's a good idea to start this section with an overall summary of your work and highlight the main findings of your research. Then you must interpret your findings clearly to your readers one by one. Then you must talk about how your results fit into existing research and literature. Then you should talk about the implications and possible benefits of your research. Then you must talk about the limitations and shortcomings of your research. And finally, you can talk about the possible future directions of your work.

Let's look at some examples. It's a good idea to start your discussion section with a summary of your work. The best way to do this will be to restate your research question and then remind your readers about your methods. And finally, providing an overall summary of your results. Look at this example here: ***“Our aims were to compare the effectiveness and user-friendliness of different storm detection software for storm tracking. On the basis of these aims, we ran multiple experiments with the same conditions using different storm detection software. Our results showed that in both speed and accuracy data, ‘software A’ performed better than ‘software B’.*** This example is from an engineering research paper. The authors are restating their aims first, which is to compare different types of storm tracking software. Then they are providing a brief summary of the methods. Here they are testing different storm tracking software under different conditions to see which performs the best. Then they are finally providing the main finding, which is that they found ‘software A’ a better than ‘software B’. This is a very good example of how to start the discussion section by presenting a summary of your work.

Here's another example: ***“The section presents and discusses the results of the experiment according to the research hypothesis. H1 proposes that instructional methods and gender influence the achievements of students. A three-way ANKOV A was used to test the hypothesis. The test reveals that there are statistically significant differences attributed to the instructional method variable in favor of project-based learning.***” The authors are restating the hypothesis in the first line. They are saying that the aim of their work was to test the hypothesis that the type of learning method influences student achievement and grades. In the second line they are providing information about the statistical tests they used to test the hypothesis. Finally, in the last line, they are providing their main finding. They are saying that they found that project-based learning has a significant positive influence on student grades. As you can see this paragraph pretty much summarizes the entire

paper in three lines. The next step is to interpret your results. You have to explain your results clearly to your readers.

Here is an example. ***“The results of this study indicate significant differences between classical music and pop music in terms of their effects on memory recall and cognition. This implies that as the complexity of the music increases, so does its ability to facilitate cognitive processing. This finding aligns with a well-known “Mozart effect”, which suggests that listening to classical music can enhance cognitive function.”*** This is a study that investigates the impact of different types of music on people. The authors are saying that their results show that there is a significant difference between pop music and classical music in terms of memory recall and cognition. Now they are providing the interpretation of the findings. They think it is because there is a link between the complexity of music and cognitive processing. They are also making reference to a well-known theory called the “Mozart effect” to back up their findings. It is a nicely written passage and the authors interpretation sounds very convincing and credible.

Now look at this example here. ***“We found the largest differences in ocean heat from 1993 to 2021 averaged about 0.16 Watts per Square Meter. The differences in ocean heat in these years can, in part, be attributed to excess heat from the earth’s surface being passed back to the ocean. This could also be due to the rising amounts of greenhouse gases preventing heat radiated from the earth’s surface from freely escaping into space.”*** The authors are providing multiple interpretations of the same finding. This passage is from a paper about climate change. The authors are saying that their results show that the largest difference in ocean heat occurred between 1993 and 2021. Now they are providing the interpretation of the result. They are presenting two possible explanations for this finding. Number one, they are saying that this might be due to heat reflecting off the ocean surface back to space. And number two, this might also be due to an increase in greenhouse gases which happened during this period. The authors believe that both interpretations are plausible and have explained them clearly in the text.

The next step is to compare your results to the literature. You have to explain clearly how your findings compare with similar findings made by other researchers. Here’s an example: ***“Most importantly, we demonstrated a significant relationship between exercise and stress reduction in our sample population. This finding is in line with most studies [1-2] investigating the relationship between physical activity and stress reduction, which have consistently found a positive correlation. The present study, therefore, extends these findings.”*** The authors are saying that they found a link between

exercise and stress reduction. Then they are saying that their findings are in agreement with most studies in the literature published on this topic. So, their findings provide further evidence of the link between exercise and stress reduction. As you can see, the authors have clearly explained how their results fit into existing research on the topic.

Here is another example where authors are providing details of papers in the literature that both support and oppose their findings: ***“Our analysis predicts that climate change will have a significant impact on wheat yield. This finding undermines one of the central pieces of evidence in some previous simulation studies [1-3] that suggest a negative effect of climate change on wheat yield, but the result is entirely consistent with the predictions of other research [4-5] that suggests the overall change in climate could result in increases in wheat yield.”*** The authors are saying that their results show that climate change will have a significant effect on the wheat production. Then they are saying that there are some papers in the literature that are in agreement with their findings. However, there are also many papers in the literature that disagree with their findings. This is very important. Your discussion should be two sided, not one sided. You should not ignore the literature that does not corroborate your findings. The next step is to explain to your readers how your findings will benefit society and the research community. You have to clearly explain the value of your work to your readers.

Here is an example. In this paper the authors are saying that their results show a strong relationship between childhood education and future career success: ***“First and foremost, results indicate that there is a strong relationship between early childhood education and future career success. The findings of this study have clear applications for policymakers and educators. This research adds to existing literature that supports the idea that investing in early education can have long-term benefits for individuals and society as a whole.”*** Then they are talking about the implications of their study. The authors are saying that their findings will help policymakers and educators make better decisions in the future. They’re also saying that their findings reemphasize the idea that investing in early education will have long term benefits for both individuals and society. As you can see, the focus here is on the broader impact of the study.

Here is another example: ***“The results contribute insights with regard to the management of wildfire events using artificial intelligence. One could easily argue that the obvious practical implication of this study is that it proposes utilizing cloud-based machine vision to detect wild fires in real time, even before the first responders receive emergency calls”***. In this paper

the authors are saying that their findings indicate that AI can be used to effectively manage wildfire events. Then they are talking about the practical implications of their study. They are saying that their work has proven that machine learning can be used to detect wildfires in real time. This is a great practical implication and can save thousands of lives. As you can see, after reading the passage, you can immediately understand the value and significance of the work. It is very important that you discuss the limitations of your study. Limitations are flaws and shortcomings of your study. You have to tell your readers how your limitations might influence the outcomes and conclusions of your research. Most studies will have some form of limitation, so be honest and don't hide your limitations. In reality your readers and reviewers will be impressed with your paper if you are upfront about your limitations.

Look at this example: ***“Study design and small sample size are important limitations. This could have led to an overestimation of the effect. Future research should reconfirm these findings by conducting larger-scale studies.”*** The authors are saying that the main limitations of the study are the small sample size and weak study design. Then they explain how this might have affected their results. They are saying that it is possible that they are overestimating the actual effect they are measuring. Then finally, they are telling the readers that more studies with larger sample sizes should be conducted to reconfirm their findings. As you can see, the authors are clearly explaining three things here. Number one, what is the limitation? Number two, how it might affect the study outcomes? And number three, what should be done to address the limitation?

Here is another example: ***“However, some study limitation should be acknowledged. The experiments do not fully consider the problems that can appear in real situations. Hence, caution should be taken with generalizing the findings and applying them to real-life situations.”*** The authors acknowledge that their study has some limitations. Then they explain what the limitations are. They are saying that their experiments do not consider all problems that might occur in real-life situations. Then they explain how this might affect their research outcomes. They are saying that readers should be careful when generalizing the results to practical real-world situations because there is a possibility that the methods might fail. It is important to remember not to end your paper with limitations. Finish your paper on a positive note by telling your readers about the benefits of your research and possible future directions.

Here is an example. The authors are starting the final paragraph of the discussion section by highlighting the benefit of their work which is the use

of biomass as a renewable source of energy: ***“While this study highlights useful insights about the potential of biomass as a renewable energy source, future research can extend this research in several ways, including research on how to tackle challenges that hinder the sustainability of renewable energy sources towards climate change mitigation, such as market failures, lack of information and access to raw materials”***. Then they talk about future research. They are saying that future research can focus on how to improve the sustainability of biomass production. As you can see, this is a very nice way to end the paper.

How to Write the Conclusion Section of a Research Paper?

In this video we will learn how to write the Conclusion section of your research paper. The conclusion is the last section of your research paper. This is the place to provide the final take home message to your readers. Some journals require a separate conclusion section. Whereas, in some journals the last paragraph of the discussion section is kind of treated as a conclusion. This section typically contains the following components. You can start by reminding your readers about the aims and objectives of your work. You must keep it brief, and preferably in one or two sentences. Then you can highlight the overall findings of your work and explain what we have learned from your study. Then you can talk about the importance and significance of your findings. Finally, you can finish up by talking about the future implications and possible applications of your work.

Here is an example: ***“In summary, the use of natural light in office buildings is investigated in this study. The results show that there is a strong correlation between the use of natural light and the productivity of employees. The results demonstrated in this work provide a new perspective on the importance of natural light in office buildings and have important implications for the design and construction of future office buildings.”*** The authors are starting the conclusion paragraph with a single line summary of their work. They are saying that the objective of their research was to investigate the effect of natural light in office buildings. Then they are explaining what they have learned from their work. They are saying that they have found a link between natural light and the productivity of employees. They are explaining the future applications of their findings. They are saying that the results of this study will be helpful in the design and construction of future office buildings.

Here is another example: ***“In summary, this study provided evidence for the impact of climate change on sea level. Specifically, research showed***

that the sea level has risen by an average of 8 inches over the last century, and the rate of sea level rise has accelerated in recent decades. To our knowledge, this is one of the most comprehensive studies to date assessing the impact of climate change on sea level. Therefore, these findings have important implications for coastal communities and infrastructure planning.” The authors are starting the conclusion passage with the major findings of their research. They are saying that their study has provided evidence that there is a link between climate change and sea level rise. Then they are explaining the significance and importance of their findings. They are saying that it is one of the most comprehensive studies done on this topic to date. Then they are finishing the passage with the future implications of their work. They are saying that their findings will help coastal communities with infrastructure planning in the future.

Here is a slightly different example: *“Our objective was to test for allergies in patients with titanium dental implants. The major contribution of this study can be summarized as follows: (1) Identifying the associated symptoms of titanium allergies; (2) Various tests for detecting titanium allergies and (3) Highlighting the best ways to manage titanium allergies. The results of this study are very important for the clinical management of patients with titanium allergy.”* The authors are starting the conclusion paragraph with the study objective, which is to test for allergies in patients with titanium dental implants. Then they are listing the contributions of their work. They are saying they have made three contributions. Number one, they have shown how to identify the symptoms of titanium allergy. Number two, how to conduct tests to confirm the allergy. And number three, how to manage the allergies in the long term. Then they are finishing the paper off with some information about the future implications of their work. They are saying that the findings of their work are very important for the clinical management of patients with titanium allergy.

Unit 6. How to Write the Abstract Section of a Research Paper?

Hi, this is Chelsea from Ref-n-Write. In this video we will learn how write the abstract section of your research paper. The abstract is one of the most important sections of your research paper. People normally see your abstract first because it will be available in conference programs and various databases. This section should be the last thing you write because this is a summary of entire paper along with your final conclusions. **The abstract is like a mini paper.** The abstract should be self-sufficient, and just by reading the abstract, the reader should get a rough idea of what you have done in the

paper. Here are three reasons why you should spend a decent amount of time writing your abstract. 1) The abstracts are available for free. Readers will read the abstract first and then decide whether to purchase the full version of the paper or not. 2) After submitting your paper to a journal, the editor reads the abstract and decides whether to send your paper for review or to reject it. 3) The reviewers read the abstract first, if it is badly written, then they won't bother wasting time reading the rest of the paper. They will assume that the rest of the paper will be equally bad and will reject the paper. So, it is very important that the readers have a positive reaction to your abstract.

The abstract should clearly answer the following questions. 1) Why you are writing this paper. 2) What did you do? 3) What are your results? 4) What are the implications of your findings?

There are two types of abstracts: structured abstract and unstructured abstract. Most journals follow the unstructured format. This is basically writing your abstract as a continuous paragraph. However, some journals follow a structured format. You are expected to structure your abstract under different headings. Let's look at an example abstract that is written in both structured and unstructured formats. This is an abstract from a medical research paper: ***“Obesity and vitamin D deficiency are two conditions that usually occur simultaneously in patients. However, it is unclear whether vitamin D supplements aid in weight loss among obese patients. In a randomized, placebo-controlled trial, we examined the effect of vitamin D supplements on weight loss over 24 weeks in 100 obese women with BMI as a primary outcome. Mean BMI decreased in the vitamin D group (22.3 ± 0.2) compared to the placebo group (26.02 ± 0.3), and a significant effect was observed ($p \leq 0.001$). The findings have proven that vitamin D supplementation can be used as a potential therapeutic option for weight loss programs.”*** The authors are starting with a topic and the problem definition. They are saying that vitamin D and obesity appear to be linked, and it is unclear if vitamin D supplements will contribute to weight loss among obese patients. Then they are explaining the method. They conducted a randomized clinical trial with 100 participants. The trial went on for 24 weeks and they measured BMI during the study. Then they move on to the results. They are saying that they found a significant difference in BMI between the vitamin D group and the placebo group. Finally, they talk about the implications of their study. They are saying that their study has provided evidence that vitamin D can be used as a therapeutic option for weight loss programs. The abstract nicely summarizes the essence of the paper. After reading it, you know exactly what to expect from the paper.

Now, let's look at the same example written in a structured format:
“Background and objectives: *Obesity and vitamin D deficiency are two conditions that usually occur simultaneously in patients. However, it is unclear, whether vitamin D supplements aid in weight loss among obese patients.*

Methods: *In a randomized placebo-controlled trial, we examined the effect of vitamin D supplements on weight loss over 24 weeks in 100 obese women with BMI as a primary outcome.*

Results: *Mean BMI decreased in the vitamin D group (22.3 ± 0.2) compared to the placebo group (26.02 ± 0.3), and a significant effect was observed ($p < 0.001$).*

Conclusions: *The findings have proven that vitamin D supplementation can be used as a potential therapeutic option for weight loss programs.”*

The abstract is structured under four headings. 1) **Background and objectives**, 2) **Methods**, 3) **Results**, 4) **Conclusions**. Please, check your journal requirements before you start writing your abstract. As many journals in the field of health sciences require a structured abstract.

Here is another example: *“The quality of video is affected by errors occurring at various stages of the production chain and are very challenging to fix. Automatic error correction methods based on the principles of motion flow estimation and image correlation are proposed. The new algorithms outperform existing techniques with superior performance in quality and speed. The new technique eliminates the visual inspection process and automatically restores the video integrity.”* This example is from an engineering research paper. The authors are starting with the topic. They are saying that the topic relates to removing errors from digital video. Then they are explaining the method. The authors are saying that they have proposed a new technique that can fix these errors automatically. Then they are presenting the results. They are saying that their methods are faster and superior to existing methods. Finally, they are finishing up with the implications of the study. They are saying that their methods will remove the need for the painstaking process of visually inspecting the video for errors. As you can see this abstract follows the standard format of topic, methods, results and implications.

Appendix

Style Guide Overview

https://owl.purdue.edu/owl/avoiding_plagiarism/guide_overview%20.html

This resource provides a summative overview of style guides. Students and instructors will find this resource particularly useful for understanding what a style guide is, why and how style guides are used, and suggestions for exploring style guides across the disciplines.

What is a style guide? Why & how do we use them?

Most people know style guides as rules for citation formatting. Common and popular style guides include MLA and APA, both of which the OWL maintains extensive resources on, but there are many more that the OWL does not cover in depth or at all.

Style guides, however, are not just for citation. They include a wide range of rules and guidelines for works in their respective fields, from grammar and language use to the font and size of headings in a work. Generally, style manuals include everything a writer needs to know in order to make their work look and read just like every other work written in that style – the look of the page, the way other authors are referenced in the body of the work, and even the tone of the writing. In the corporate sphere, "style guide" is often synonymous with "branding guide," and in a way academic style guides can be thought of similarly. APA style brands papers with specific guidelines in the same way Apple's style guide brands their websites, advertisements, products, documentation, and even stores: you know an APA paper when you see one, just like you know an Apple device when you see one.

In other words, style guides are used as a way of making common elements consistent across documents written by many writers, in many places, and in many circumstances; as a result, readers from any university (or other audience groups) can read a paper written in APA style and know immediately how to navigate the headings of the paper, which details will be listed

in the abstract, how quotes will be introduced and marked, where to look for important citation information, and what each citation element represents.

There are a great deal of style guides, both in academia and in industry; the OWL has resources on some of the more common ones, but not every style. Some general style guide examples include:

Common Academic & Professional Style Guides & their Primary Field of Usage

Style	Usage
Associated Press Style (<u>AP</u>)	used by journalists and other news & media writers
Chicago (sometimes written Chicago/Turabian, or <u>CMoS</u>)	used in the humanities
Institute of Electrical & Electronics Engineers Style (<u>IEEE</u>)	used in engineering & computer science
American Medical Association (<u>AMA</u>)	used in nursing & other medical fields
Association for Computing Machinery (<u>ACM</u>)	used in computing and information technology fields
The American Society of Mechanical Engineers (<u>ASME</u>)	used in mechanical engineering

For more information use site <https://www.scribbr.com/citing-sources/citation-styles/>

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